

Book 1

Mastering Thinking Skills

Selective and Scholarship Tests

- ✓ Enhances higher-order and critical thinking skills
- ✓ Improves knowledge and decision-making in problem solving
- ✓ Develops mental, cognitive and strategic abilities

Yvonne Kang

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Mastering **Thinking Skills**

Selective and Scholarship Tests

- ✓ Enhances higher-order and critical thinking skills
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**A
FIVE SENSES
PUBLICATION**

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Instructions

Instructions for Candidates

- There are 4 Thinking Skills tests in this book. There are 40 questions in each test. The time limit for each test is 40 minutes.
- For each question there are four possible answers, A, B, C and D. Choose the one correct answer and record your choice.
- You will not lose marks for incorrect answers, so you should attempt all 40 questions.
- You must complete each test within the time limit. There will not be any extra time at the end of the test to review your answers.
- Calculators and dictionaries are NOT allowed.

Test 1

1. Harry wants to buy ice cream from the school canteen. As well as considering the cost, he also considered the taste and popularity of the ice cream.

If the ice cream is cheap, then it only needs to be tasty. If the ice cream is expensive, then it either has to be extremely tasty and of average popularity or of average taste and extremely popular.

Harry wants vanilla ice cream, which is expensive, but he did not buy it.

What must have been the reason?

- A It was of average taste and extremely popular.
 - B It was of average taste and of average popularity.
 - C It was extremely tasty and of average popularity.
 - D It was extremely tasty and the most popular.
2. In a survey of television viewers, everyone who liked movies liked animations. Everyone who liked animations liked cartoons, but no one who liked animations liked reality shows. David, Amanda, Brian and Julie all took part in the survey.

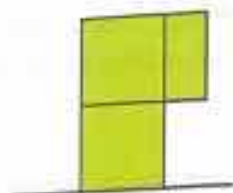
Based on the above information, which one of the following must be true?

- A If David likes cartoons, he also likes animations.
- B If Amanda likes movies, she does not like reality shows.
- C If Brian does not like movies, he does not like animations.
- D If Julie does not like reality shows, she does not like cartoons.

3. A sculpture in the centre of the local park is made from three identical cubes. The views from the north and east are shown below.



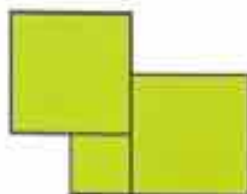
from the north



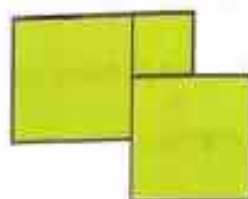
from the east

What does the sculpture look like from above?

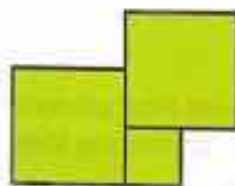
A



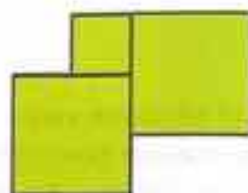
B



C



D



4. A factory worker has five children. No one else in the factory has five children.

Which one of these sentences can be concluded from the above information?

- A All workers in the factory have five children each.
- B Everybody in the factory has children.
- C Some of the factory workers have more than five children.
- D Only one worker in the factory has exactly five children.

5. Students at school have been sick all winter long. Many students have missed over a week of school due to the viruses passing from student to student.

A few students, though, have not been sick at all. They have had perfect attendance.

When asked why they had not been sick, the students explained that they cleaned their hands regularly with hand sanitiser.

What is the cause of the students not getting sick?

- A They missed no school.
- B They missed many days of school.
- C They used hand sanitiser.
- D They ate breakfast every day.

6. Paul, Jimmy and Neil go swimming regularly. Paul goes on Saturday, Monday and Wednesday. Jimmy goes every second day, while Neil can only go once a week.

If they were all there on Wednesday, what is the minimum number of days before the three boys meet up again?

- A 3
- B 5
- C 8
- D 10

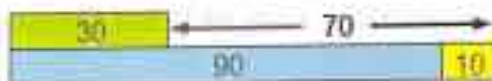
7. The children Kerry, Sam, Ethan, Lee and Susan are all in one family. Lee is 1 year younger than Ethan. Sam and Ethan are twins. Kerry is 3 years older than Lee. Sam is 9 years old. Susan is older than Ethan, but younger than Kerry. Only one of the following sentences is true.

Which one is it?

- A Ethan is 11.
B Kerry is 2 years older than Susan.
C All the children are 9 or older.
D Kerry is 11.

8. Ancient builders built their houses with the aid of three sacred measuring sticks, 10 cm, 30 cm and 90 cm long. With these, laid down together in various combinations, and using none twice, they could measure any unit distance from 10 to 130 cm.

For example, the method of measuring 70 cm is shown below:



Chris claims he can measure anything from 10 to 140 cm if his sticks are 10, 30 and 100 cm long. However, there is a distance he cannot measure.

What is the distance that cannot be measured?

- A 20 cm
B 50 cm
C 60 cm
D 80 cm

9. Over a year a school monitored the attendance record of its 8 students and recorded the following information on the number of days when there was just one person absent, two people absent and so on.

Number of absentees	Days
1	40
2	10
3	3
4	2
5	1
6 or more	0

Absences of each individual were also recorded, but the details for student Harry were lost.

Student	Days absent
Alina	10
Ben	6
Chris	16
David	12
Ethan	20
Fred	0
George	2
Harry	?

On how many days was Harry absent?

- A 0
- B 6
- C 10
- D 16

10. I am hiding in one of three rooms. On the door of each room I have written two statements, some true, some false. However, no more than one statement on each door is false.

Room 1	Room 2	Room 3
My name is Alice	I am not in this room	My name is Peter
I am not in this room	I am in Room 1 or 3	I am not in Room 1

Which one of the following statements must be true?

- A My name is Alice.
 - B I am not in Room 1.
 - C I am in Room 1 or 3.
 - D My name is Peter.
11. William bought some lollies for 30 cents and some other lollies for 50 cents, spending a total of \$5.50.

He bought three more 50 cent lollies than those costing 30 cents.

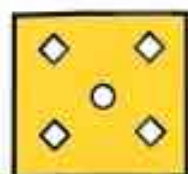
How many 50 cent lollies did William purchase?

- A 7
- B 8
- C 9
- D 10

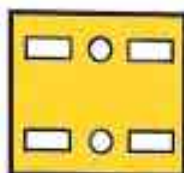
12. A piece of square paper is folded and holes are punched as shown below.



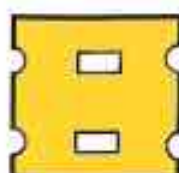
Which of the following figures indicate how it will appear when opened?



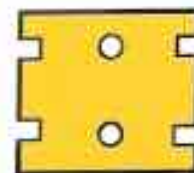
A



B



C



D

13. There are three siblings who all play different sports and different instruments. All of them are at a different age. We know that:

- Brian is the youngest
- The person who swims is the middle child
- Abby plays violin
- The oldest child plays trumpet
- The person who plays soccer is the youngest
- The person who swims is younger than the person who plays football, but is older than the person who plays soccer
- Charlie plays football
- The oldest child plays football

Who does swimming as a sport?

- A Abby
- B Brian
- C Charlie
- D Cannot be determined

14. It is said that excessive computer usage causes eye development issues in young children, aged 2-5. It is then recommended that these children stay off the computer as much as possible. It is also known that computer screens emit harmful blue light.

Which of the following best **strengthens** the argument?

- A Children learn a lot from using the computer.
- B Young children's eyes are still developing, and the blue light delays their development.
- C Children can supplement their learning experiences through using the computer.
- D Children do not know how to use the computer.

15. Lance, Steve, Trish and Vicky deliver pizzas for a pizza shop. When they deliver more than 20 pizzas in a week, they earn bonus money. Today, Lance was paid a bonus and Vicky had delivered 32 pizzas for the week.

If the above statements are true, which one the following statements must definitely be true?

- A Lance delivered 50 pizzas in the week.
- B Steve was not paid a bonus.
- C Trish delivered less than 50 pizzas.
- D Vicky was paid a bonus.

16. There are 24 students in 6F. In 6S, there are 13 boys and 11 girls.

When the classes are combined, there are 2 more girls than boys.

How many girls are there in 6F?

- A 13
- B 14
- C 15
- D 16

17. There are some cubes in a bag. The number of cubes in the bag is more than 5 but less than 15.

The cubes are either red (R) or black (B).

The teacher says:

If you take a cube at random out of the bag, the probability that it will be red is $\frac{1}{5}$.



A pupil takes one cube out of the bag.
It is red.

What is the probability of choosing a black cube from the bag next time?

- A $\frac{4}{5}$
- B $\frac{3}{5}$
- C $\frac{4}{9}$
- D $\frac{8}{9}$

18. James is a taxi driver who is considering working only on Fridays, doing a 12-hour shift from 08:00 to 20:00. He is permitted to collect passengers from the railway station only. He collects a passenger, drives them to their destination, and returns to the station to collect the next passenger.

For each journey James plans to charge a fixed fare of \$2, plus \$1 per kilometre of journey distance from the railway station to the passenger's destination. His fuel costs are \$0.40 per kilometre that he drives his taxi (including the return journey to the station). His profit for that passenger is the difference between the fare he charges and his fuel costs.

How much profit does he earn for taking a passenger to a destination 15 km away and then returning to the station?

- A \$5
- B \$10
- C \$31
- D \$14

19. Targeting is used throughout the advertising industry to make sure that ads and messages reach the right audience. Politicians do this as well. In the age of social media, politicians and people running for office are able to target their campaigns. If a candidate wants to address the concerns of women, college students, retired people or any other group of voters, they can now tailor their messages. For this reason, Danielle believes that social media should be banned during elections.

Which one of these statements, if true, most **strengthens** Danielle's argument?

- A Senators use social media to reach out to their voters.
- B Social media allows voters to stay educated about elections.
- C People change their minds depending on what advertisements they see.
- D Some politicians struggle to use technology and social media.

20. Buying a dog, as opposed to adopting one, is a much better experience. There is something very special about raising your own dog from a tiny puppy. And it's what a lot of people dream of, when they think of getting a dog. There are simply more puppies available when you purchase from a store. Buying a cute pet that you actually want will then motivate you to care for and keep the pet, rather than returning it to the shelter or abandoning it, as some owners do.

Which one of these statements, if true, most **weakens** the above argument?

- A Animals in shelters have been abandoned by owners or rescued.
- B Getting a pet has many unexpected responsibilities.
- C All puppies will grow up and people may not want them anymore.
- D Adopting a dog is a very thorough process.

21. There are 2 ways to qualify for the Maths Olympiad Competition: by getting 'A' in the most recent school report or by getting 100% in the most recent test. This year, 8 students from Sophie's school have qualified for the Maths Olympiad Competition.

Sophie: "I know that 5 students got 100% in the most recent test. That means that 3 students got an 'A' in their most recent school report."

Which one of the following sentences shows the mistake Sophie has made?

- A Some of the students from Sophie's school may have gotten 100% in the most recent test but did not get 'A' in their report.
- B There are more students who qualified this year compared to last year.
- C There are less students who qualified this year compared to last year.
- D Some of the students from Sophie's school may have gotten both 'A' in their report and scored 100% in the test.

22. People do not eat certain foods because they are picky eaters. This is especially demonstrated in our present world, due to the large variety of foods that can now be found in a large selection of stores, cafes and restaurants.

Which of the following best **weakens** the above argument?

- A People do not eat certain foods because they do not wish to.
 - B People eat whatever they want because they want to.
 - C People do not consume food at all.
 - D People do not eat certain foods due to allergies that can harm them.
23. The International Morse Code uses dots and dashes (symbols) to represent letters. These are separated by gaps.

There are two letters using only one symbol: E and T.

A, I, M and N use two;

D, G, K, O, R, S, U and W use three;

and the rest of the letters use four.

The following combinations are not used.

• — • —
 — — — •
 — — — —
 • • — —

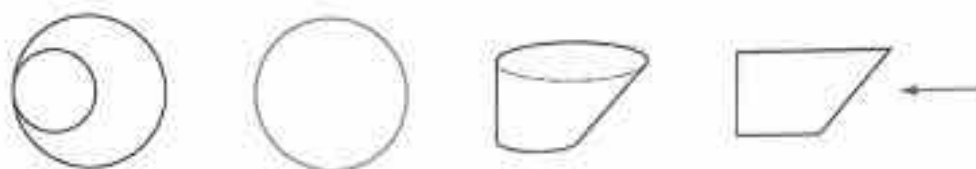
A secret message with various letters was written down, unfortunately without the spaces between the letters:

• — • • — • • • — — — • • — • • • • — •

What is the minimum number of letters that this message could have?

- A 5
- B 6
- C 7
- D 9

24. The diagrams below illustrate four views of a solid figure.

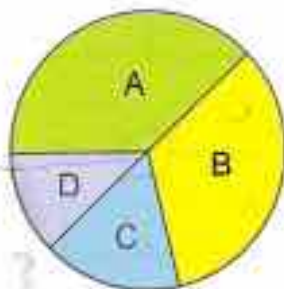


Which of the responses below illustrates the view of the figure from the direction shown by the arrow on the fourth diagram?



25. There are four political parties in the country of Quest. They are the A Party, the B Party, the C Party and the D Party.

This pie chart illustrates the distribution of the parties in the current Quest Parliament:



There are 120 seats in the Quest Parliament.

Which of the following is the correct number of seats for each of the parties?

	A Party	B Party	C Party	D Party
A	46	43	18	13
B	48	44	16	12
C	46	42	18	14
D	48	43	17	12

26. Brian is a cashier. He needs to give change for a customer, and notices that there are a different number of coins left.

<i>Coins</i>	<i>Number left</i>
50c	2
\$1	4
\$2	3

Brian needs to give \$5.00 in change. How many ways can he do this?

- A 5
- B 6
- C 7
- D 8

27. I was in charge of allocating the budget for an office party, but I accidentally mixed up the data. The table below shows the incorrect amounts next to all of the categories, except for the cake, of which I'm certain has the right amount listed next to it.

Category	Budget (\$)
Drinks	25
Ice Sculpture	130
Balloons	115
Food	70

I recall that the combined amount for the drinks and ice sculpture was less than the combined amount for balloons and food. Also the combined amount for the drinks and balloons was less than the combined amount for the ice sculpture and the food.

Which one of these statements cannot be true?

- A The budget for balloons was less than the budget for food.
- B The budget for food was the highest of all the items.
- C The budget for food was less than the budget for drinks.
- D The budget for balloons was less than the budget for the ice sculpture.

28. Many business offices are located in buildings having two to eight floors. If a building has more than three floors, it has a lift.

Which one of these sentences can be concluded from the above information?

- A All floors may be reached by lifts.
- B Only floors above the third floor have lifts.
- C Seventh floors have lifts.
- D Second floors do not have lifts.

29. The amount of a certain medicine required by a baby depends on its weight, as shown in the table below.

Weight (kg)	Amount of medicine (ml)
2.5 – 3.5	2.9
3.5 – 5.0	3.8
5.0 – 6.5	4.3
6.5 – 8.0	4.5

Four babies were weighed and given the correct amount of medicine in October and again in January.

Which baby required the greatest increase in the amount of medicine from October to January?

	Weight (kg)	Weight in January (kg)
A	5.2	7.1
B	2.8	3.9
C	3.8	6.6
D	6.7	7.3

30. Twelve contestants are taking part in a series of challenges. Today's challenge was an obstacle course, which they completed against the clock. The contestant who completed the course in the shortest time was awarded 5 points, the contestant with the second-best time was awarded 2 points, and contestants whose times were within 90 seconds of the second-best time were awarded 1 point each.

The order in which the contestants competed and their times (min:sec) are shown below.

Arthur 12:33	Lesley 10:51	Sana 12:26
Tammy 12:57	Vicky 12:23	Carol 12:08
George 11:39	Linda 13:25	Adam 13:04
Callum 13:34	Scott 13:18	Pippa 12:45

How many of the contestants scored points in today's challenge?

- A 3
- B 5
- C 7
- D 9

31. The prices for passes for a tourist bus are shown in the table below.

Pass for 1 day	Pass for 4 consecutive days	Pass for 7 consecutive days
\$6	\$11	\$22

Sandy needs a pass on the following days of the month: 2, 3, 4, 6, 10, 11, 12, 14, 15, 16.

How much is the cheapest valid combination she could use?

- A \$33
- B \$39
- C \$44
- D \$46

32. Ten students are studying English. They take three tests, each marked out of 50. The results of the first two tests taken are shown below.

	Test 1	Test 2
Student A	24	27
Student B	42	43
Student C	30	34
Student D	37	27
Student E	33	30
Student F	34	39
Student G	16	26
Student H	44	22
Student I	50	27
Student J	35	39

A prize is awarded to the student with the highest combined total score from the three tests. If two students are tied for the highest score, nobody gets the prize.

As the score currently stands, one student is able to be sure to win this prize as long as a high enough score is achieved in the final test.

What is the minimum score in the final test that will guarantee that student gets the prize?

- A 42
- B 43
- C 46
- D 50

33. Whenever the light on the side of Fred's computer is flashing, it means that the battery is going to run out soon.

Fred: "The light is not flashing right now, so the battery will last a long time."

George: "No, that's not right. The light is on continuously – that must mean that the battery has already run out!"

If the information in the box is true, whose reasoning is correct?

- A Fred only
 - B George only
 - C Both Fred and George
 - D Neither Fred nor George
34. All Labrador retrievers bark a great deal. All Saint Bernards bark infrequently. Each of Rosa's dogs is a cross between a Labrador retriever and a Saint Bernard.

Rosa: My dogs are moderate barkers.

Which one of the following sentences shows the mistake Rosa has made?

- A The mix of a great barker and quiet dog is a moderate barker.
- B Rosa prefers a moderate barker.
- C Rosa lives in an apartment so she must have a moderate barking dog.
- D There is no information on barking of mixed dogs.

35. I buy my sandwiches each day from the school canteen. The prices are shown in the table below.

Filling	Small roll	Large roll	Baguette
Cheese	\$1.00	\$1.20	\$1.50
Egg	\$1.20	\$1.50	\$1.80
Beef	\$1.80	\$2.00	\$2.20
Chicken	\$1.80	\$2.00	\$2.50
Ham	\$2.00	\$2.50	\$2.80
Pork	\$2.20	\$2.50	\$2.80
For salad add	\$0.20	\$0.30	\$0.50

I plan to have a large cheese roll on Monday and Thursday, a chicken baguette with salad on Tuesday and a large egg roll on Wednesday and Friday.

The canteen has a special offer: if I order all of my sandwiches for the week then I will get the cheapest one free.

How much will I need to pay for my sandwiches for the week?

- A \$9.00
 B \$7.20
 C \$7.80
 D \$8.40
36. If Tiffany does not set an alarm, she will wake up late.
 If she wakes up late, she will miss the bus.
 If she does not miss the bus, she may get to work on time. If not, she will be late.
- If the above statements are correct, which one of the following is not possible?
- A Tiffany sets an alarm but wakes up late.
 B Tiffany catches the bus and is late to work.
 C Tiffany wakes up early and misses the bus.
 D Tiffany wakes up late but is not late for work.

37. Amy writes down the whole alphabet, in order.

She then chooses a pair of letters, A and M. She finds the letter that is three quarters of the way between them.

Then she chooses another pair of letters. She finds that the letter half-way between these letters is the same as the letter she found three quarters of the way in the previous pair.

If the first letter of the second pair is D, what is the other letter?

- A P
- B K
- C N
- D Q

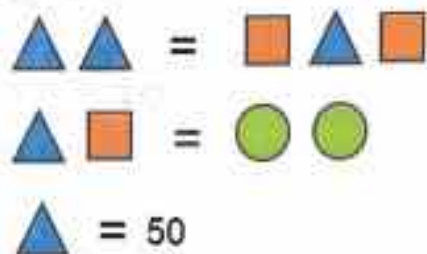
38. Ross is auditioning for the talent show 'Australia's Got Talent'. The audition process examines the different skills of each participant. All contestants needed to fill out a form to be able to pass the audition.

If a contestant could do acrobatics, they did not need to know how to sing. If they could not, then they either had to be able to sing decently and dance perfectly, or sing perfectly.

Ross is an experienced acrobat but failed to pass the audition. What must have been the reason?

- A He did not know how to sing.
- B He did not dance perfectly.
- C He performed badly on the day of the audition.
- D He did not fill in the form.

39. Consider the diagram below.



Which of the following statements is true?

- A $\bigcirc < \triangle$
- B $\square > \bigcirc$
- C $\triangle < \square$
- D $\triangle < \bigcirc$

40. Although computer game sales have increased steadily over the past 3 years, we can expect a reversal of this trend in the very near future. Historically, over three quarters of computer games sold have been purchased by people from 13 to 16 years of age, and the number of people in this age group is expected to decline steadily over the next 10 years.

Which one of the following, if true, would most **weaken** the argument?

- A Most people 17 years old or older have never purchased a computer game.
- B New technology will undoubtedly make entirely new entertainment options available over the next 10 years.
- C The number of different types of computer games available is unlikely to decrease in the near future.
- D Most of the people who enjoyed playing computer games will continue to purchase them even if they become older.

1. Fiona believes that eyeglasses offer many benefits over contact lenses. She says that they require very little cleaning and maintenance, you don't need to touch your eyes to wear them (decreasing the risk of eye infections), and glasses are cheaper than contact lenses in the long run since they don't need to be replaced as often.

Which one of these statements, if true, most **strengthens** Fiona's argument?

- A Some types of contact lenses are reusable.
 - B Glasses come in many different styles and colours.
 - C Some eye infections can lead to blindness.
 - D Contact lenses need to be kept in solution.
2. James is choosing between dog breeds to get as his first pet. Since James has a fur allergy, he needs to consider their shedding levels as well as conduct a cuteness test and assess their liveliness.

If a dog does not shed a lot, then they only need a low cuteness level and an average liveliness. If a dog sheds a lot, then they have to be very cute and very lively.

The first dog that caught James' eye is a poodle, which James thought was extremely cute, but James did not choose it.

What must have been the reason?

- A The poodle does not shed a lot.
- B The poodle sheds a lot.
- C The poodle does not shed a lot and is not very lively.
- D The poodle sheds a lot and is very lively.

3. I have three numbers following a certain rule.

The first number has two digits.

The second number is the first number plus the sum of the digits of the first number.

The third number is the second number plus the sum of the digits of the second number.

The third number is 44.

What is the first number?

A 25

B 22

C 20

D 29

4. In a survey of school children, everyone who liked drawing liked reading.

Everyone who liked reading liked singing, but no one who liked reading liked dancing.

Steve, Nicole, Kevin and Sarah all took part in the survey.

Based on the above information, which one of the following must be true?

A If Steve likes singing, he also likes reading.

B If Nicole does not like dancing, she does not like singing.

C If Kevin does not like drawing, he does not like reading.

D If Sarah likes drawing, she does not like dancing.

5. When it is cold, bears tend to be lazier. This is an observed trait of every bear throughout the year, and it is thought to be because of their biological features.

Which of the following best **strengthens** the above argument?

- A Bears hibernate during the winter.
- B Bears hunt a lot more during the winter.
- C Bears kill more prey during the winter.
- D Bears are only awake as long as the sun is up.

6. Ashley works for a cosmetics company. She was in charge of ordering in the new fragrances but made a mistake and mixed up the data in the table below. However, she is sure that the jasmine is listed next to its correct amount.

Fragrance	Cost (\$)
Jasmine	56
Rose	63
Fruity	29
Sandalwood	82

Ashley is sure that the combined cost of the jasmine and fruity fragrances is more than the rose and sandalwood. She is also sure that the combined cost of the jasmine and sandalwood fragrances is more than the rose and the fruity fragrance.

Which of the following statements cannot be true?

- A The fruity fragrance costs more than the rose fragrance.
- B The fruity fragrance costs more than the sandalwood fragrance.
- C The rose fragrance costs more than the jasmine fragrance.
- D The jasmine fragrance is the second cheapest fragrance.

7. Science is not important in our modern world. Instead, the world is much like it was 2000 years ago, with the traditional methods of healing being the best.

Which of the following best **weaken** the above argument?

- A Healing crystals are able to cure all forms of cancer.
 - B Science is responsible for all the modern medical developments that have ensured a greater chance of humanity's survival.
 - C Medicine has killed more people than it has healed.
 - D Humans have survived without science for a long time.
8. I live on the 6th floor of a block of apartments that has 9 floors in addition to the ground floor. The lift, unfortunately, cannot be controlled. It runs continually up from the ground floor to the 9th floor and down again, taking 4 seconds to move from one floor to the next and stopping for 12 seconds at every floor. I usually wait for the lift to take me up, but I often walk down the stairs, which I have found takes me 20 seconds from one floor to the next.

This morning as I left my apartment the lift had just left. Looking at the screen, I discovered that it was going up, so I decided to use the stairs instead.

How much sooner did I reach the ground floor than if I had taken the lift?

- A 60 seconds
- B 16 seconds
- C 24 seconds
- D 44 seconds

9. Governments do not tax aviation fuel, despite evidence that the rapid growth in air travel is causing more and more pollution, and thereby contributing to climate change. Since this is bad for everyone, airlines should be required to pay a tax on fuel. The money raised could be used to seek ways to deal with the problem of pollution. Airlines would pass on the cost by increasing airfares, thus passengers would be making a financial contribution to compensate for the damage they cause. This would be the fairest solution to the pollution problems caused by air travel.

Which of the following, if true, most **strengthens** the above argument?

- A Fewer people would fly if airfares were increased.
- B World trade would suffer if aviation fuel were taxed.
- C Scientists think that the process of climate change cannot be reversed.
- D Air travel has increased because more people travel to holiday destinations.

10. A farmer has 315 female sheep.

This year all but 12 of them have had lambs.

None of them have had more than two lambs.

350 lambs have been born in total.

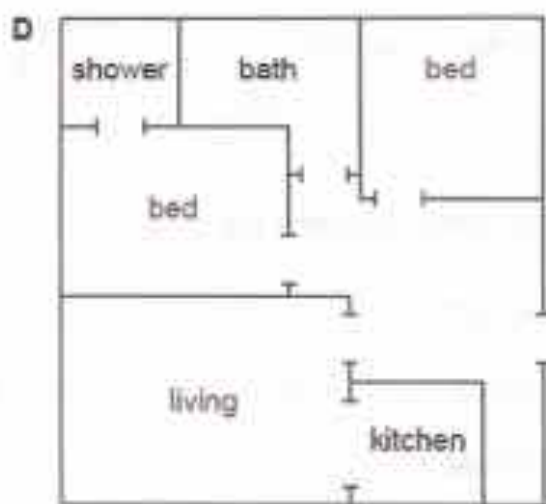
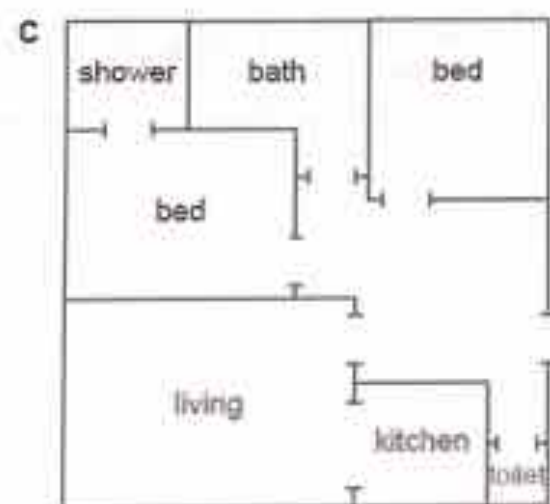
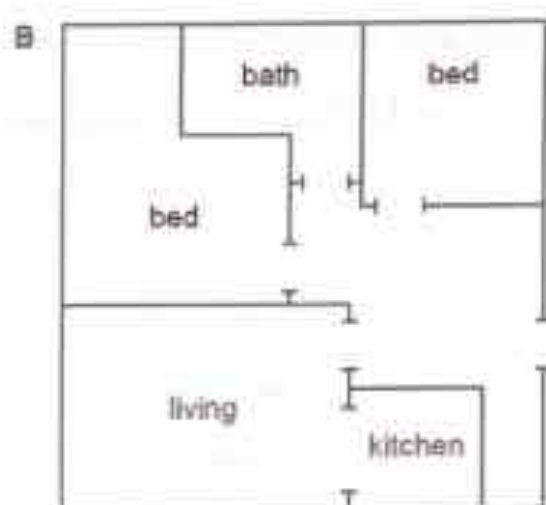
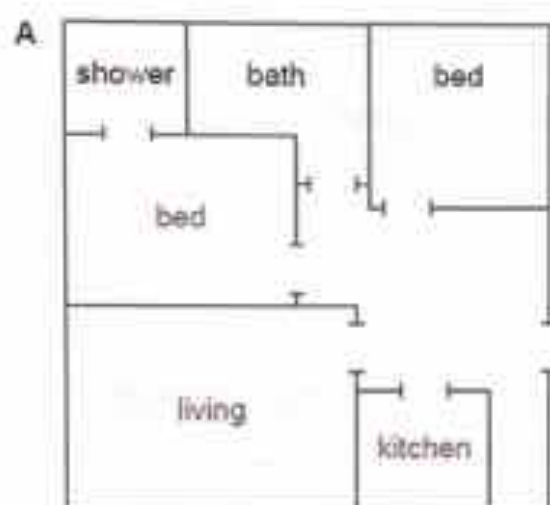
How many of the sheep have had one lamb only?

- A 112
- B 114
- C 131
- D 256

11. The following is an estate agent's description of a flat which he has for sale:

A spacious 2-bedroom apartment, well-appointed and in a good decorative order. The front door leads to an entrance hall off which there is a large living room, two bedrooms and a bathroom. The living room leads to an open plan kitchen with a good range of appliances. The main bedroom has a shower room. There is a good sized second bedroom and the bathroom is fitted with a wash/hand basin, bath and toilet. An early viewing is recommended.

Which of the following floor plans could represent the apartment described above?



12. Some employers operate a three-shift system. This means that in 3 weeks, a worker will have to work, for example, from 6 am to 2 pm in the first week, from 2 pm to 10 pm in the second week, and from 10 pm to 6 am in the third week. This makes it very difficult for workers to have a routine of eating and drinking. People working a three-shift system report a severe decline in their appetite, especially during the night-shift when they would normally be asleep. Therefore, anyone working shifts like this will therefore lose weight.

Which one of these statements, if true, most **strengthens** the above argument?

- A Employees tend to dislike working shifts.
- B People who feel less hungry generally eat less food.
- C Shift work often pays better than working days only.
- D Canteen facilities are not always available to the night shift.

13. Exactly one of these chests contains gold, but only one of the four statements is true.

Chest 1	Chest 2	Chest 3	Chest 4
The gold is here.	The gold is in Chest 1 or 4.	The gold is not here.	The gold is here.

Which chest contains the gold?

- A Chest 1
- B Chest 2
- C Chest 3
- D Chest 4

14. Julie's school wanted to determine how speaking languages affected people's intelligence.

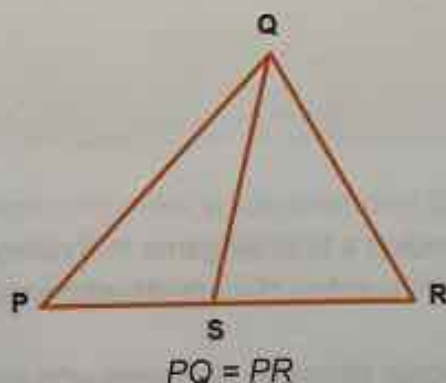
They made all the most fluent speakers take an IQ test. This is what they found:

- The students most fluent in Spanish were smarter than the students most fluent in Italian.
- All the Italian speakers were smarter than all of the Latin speakers.
- All the Latin speakers were smarter than most of those who spoke Spanish.

Which of these sentences can be concluded from the above information?

- A Spanish speakers had the highest range of intelligence.
- B People who spoke Latin were generally good at speaking Spanish.
- C On average, the smartest people could speak Spanish.
- D To be intelligent, it is better for a student to be able to speak Italian and Latin over Spanish.

15. There is a triangle PQR and it has been divided into two smaller triangles as shown below.



Using the diagram above, which one of the following statements is correct?

- A SQ is greater than PQ.
- B $PQ + RQ$ is equal to PR.
- C PS is smaller than SR.
- D The relationship cannot be determined from the information given.

16. An advertisement for a cleaning product claimed:

Quick-Stain Remover never fails to remove a stain.

Four cleaning products were used on bathroom walls, and two questions were asked in each case:

Item	Was the stain on the bathroom walls removed?	Was Quick-Stain Remover used in the cleaning?
1	(a)	Yes
2	Yes	(b)
3	(c)	No
4	No	(d)

Which two of the missing answers must be known in order to test the truth of the claim in the advertisement?

- A (a) and (b)
 B (b) and (d)
 C (c) and (d)
 D (a) and (d)

17. Among the 900 people attending a football game in Sydney, there were X number of students from school C and Y number of students who were not from school C.

Value A: The number of people attending the game who were not students.

Value B: $900 - X - Y$

Value C: $X + Y$

Which of the following statements is correct?

- A Value A is greater than B.
 B Value B is greater than C.
 C Value A and B are equal.
 D Values cannot be compared.

18. There is soccer training at the local park. The first training will be on Tuesday 1st July and the final training will be on 31st July. There will be two trainings every Monday and every Tuesday, no training on 12th, 26th and 30th July, and one training on each of the other days.

How many trainings will there be altogether?

- A 36
- B 37
- C 38
- D 39

19. Lucy wants to buy 4 books. There are 4 shops that sell the books that she wants. The shops, the prices of the books and any special offers are shown in the following table.

	Standard price of a book	Special offer
Shop A	\$16	Buy 1, get 2 nd one for half price
Shop B	\$14	Buy 3, get a 4 th one free
Shop C	\$18	Buy 4, get 20% off
Shop D	\$15	none

Lucy wants to buy all her books from the same shop and to spend as little as possible.

Which shop should Lucy buy her books from?

- A Shop A
- B Shop B
- C Shop C
- D Shop D

20. A mobile is balanced as below.

Sam and Kate are discussing what shapes can be added to maintain the balance.



Kate: "Move the shapes so all 5 circles are on the left side and both hexagons are on the right side."

Sam: "Add 2 stars to both sides."

Whose statement is correct?

- A Sam only
- B Kate only
- C Both Sam and Kate
- D Neither Sam nor Kate

21. Julie's mother has written to the school. She has suggested that, for the benefit of the children, the speed limit near school grounds should be reduced to 20 km/h instead of 40 km/h. She claims that this would reduce the chances of children being put in danger of a car accident.

Which one of these statements, if true, most **strengthens** Julie's mother's argument?

- A Parents drive faster when they are late for the school.
- B People always drive faster than the speed limit.
- C People who live close to the school have seen too many people speed.
- D Driving speed directly influences reaction time.

22. I saw 7 dog owners exercising their dogs in the dog park.

Some of the dog owners had 1 dog, and some owners had 2 dogs.

Counting the legs of the dogs and the legs of the dog owners, there was a total of 58 legs.

All dogs have 4 legs and all dog owners have 2 legs.

How many dog owners had 1 dog?

- A 1
- B 2
- C 3
- D 4

23. Space is a dangerous, unfriendly place and the aftermath is even more so. Isolated from family and friends, exposed to radiation that could increase your lifetime risk for disease, a diet high in freeze-dried food, required daily exercise to keep your muscles and bones from deteriorating, a carefully scripted high-tempo work schedule, and confined with three co-workers picked to travel with you by your boss. Transitioning from one gravity field to another is trickier than it sounds. It affects your spatial orientation, head-eye and hand-eye coordination, balance, locomotion, and you're likely to experience motion sickness.

Which one of the following, if true, most **weakens** the above argument?

- A Space exploration allows new planets to be discovered, which can solve the problem of Earth's over population.
- B Neil Armstrong suffered from severe motion sickness upon landing on the moon.
- C Travelling to space requires lot of training and preparation to prevent illness.
- D Astronauts must undergo a five-year recovery process when they return to Earth.

24. Adrian, Sarah, Issac, Iris and Lucas go on a summer day camp.

- Adrian and Issac win a tennis tournament.
- Four children go on the climbing wall.
- Sarah and Lucas learn to make origami frogs.
- Iris plays tennis after she goes on the bouncy castle.
- Sarah is tired after she goes swimming.
- Adrian and Issac are the only boys that go on the climbing wall.

Which child does the fewest activities?

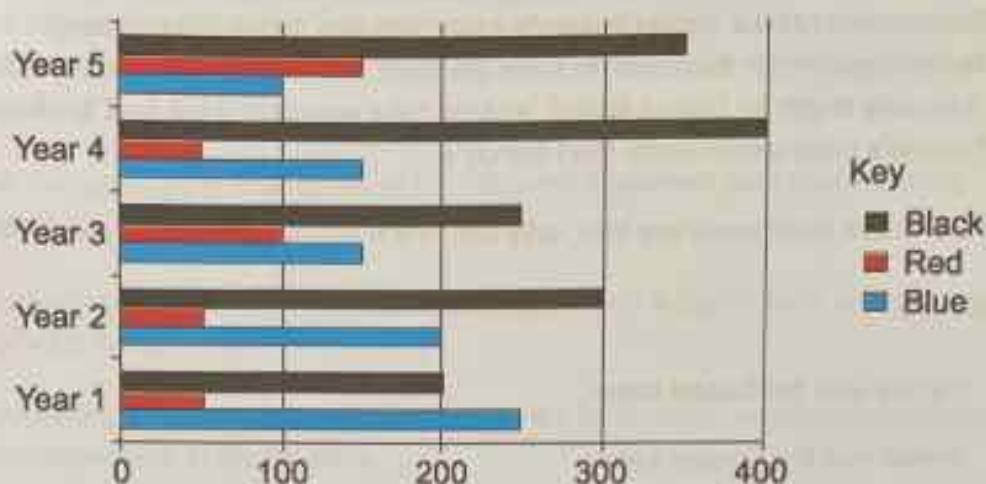
- A Sarah
- B Issac
- C Iris
- D Lucas

25. In Country X, every day is either rainy or dry. It is never rainy three days in a row and it is never dry three days in a row. It always rains on a Sunday and on a Monday and it is always dry on a Thursday.

How many dry days are there every week in country X?

- A 1
- B 2
- C 3
- D 4

26. Footlooker sells a particular type of shoe in three different colours: blue, red and black. The following graph shows the number of each colour of shoe sold in each of the last five years.



From this data for the last five years, which one of the following statements is **not** true?

- A 50% of the shoes sold in Year 1 were blue.
 - B Over half of all the shoes sold were black.
 - C The same number of shoes were sold in Year 1 as in Year 3.
 - D 450 red shoes were sold.
27. The only way to qualify for the Sports Team is if students break a record at the athletics carnival.

Amy: "I know that a total of six records were broken at the athletics carnival this year. This means that six of our students must be record-breakers!"

Which of the following sentences shows the mistake Amy has made?

- A The number of records broken this year may be higher than usual.
- B The number of records broken this year may be lower than usual.
- C Some of the students from Amy's school may have broken multiple records.
- D Some records may have been broken more than once.

28. Five bakers participated in a cake making competition where they were assessed on two criteria - the time taken to make the dish and the taste of the cake.

- Randy was faster at cooking than John, but not as fast as Frankie.
- Gordon was not the fastest baker, but his cake was ranked first in taste.
- Teresa took longer than John to bake the cake, and her cake ended up tasting worse.
- John was fourth for baking speed, and his cake was nearly the best in taste.
- Frankie's cake tasted better than Randy's.

If all the above statements are true, only one of the sentences below cannot be true.

Which one?

- A Frankie was the fastest baker.
- B Teresa was the slowest baker.
- C Frankie's cake tasted better than John's.
- D John's cake tasted better than Randy's.

29. "Whoever sprints must have both stamina and determination."

If this is true, which one of these sentences must also be true?

- A If Harry does not sprint, he cannot have stamina.
- B If Harry does not sprint, he cannot have determination.
- C If Harry does not have stamina, he cannot sprint.
- D If Harry has both stamina and determination, he sprints.

30. Everyone should consider showering at night as opposed to showering in the morning. Showering is done for the purpose of hygiene. If you shower at night, you are removing all the dirt and bacteria you may have picked up during the day. However, if you shower in the morning, it means you are not removing this dirt that has been collected during the day, but rather bringing it your bed and home when you sleep – after which will remain in your home once you shower in the morning. Using this logic, it is more hygienic to shower at night.

If the above passage is true, which of the following statement best **supports** the argument?

- A Showering is shown to relieve stress brought on by a day of work and prepares the body for sleep.
- B Showering at night requires you to turn on the light, which wastes more electricity than showering in the morning.
- C Showering is better at night if you have a lot of work to continue after a shower as it wakes you up.
- D While sleeping, sweat and gunk is still produced which needs to be washed off with a morning shower.

31. Kelly said: "Australia should adopt a new system of organ donation where everyone automatically consents, and there is an option to opt-out, rather than the current opt-in system."

Which one of these statements, if true, most **weakens** Kelly's argument?

- A There is currently a low level of organ donation and this would increase rates.
- B In a voluntary system, too few people actively decide to register.
- C This system geared towards efficiency inappropriately commodifies human organs.
- D It is unfair to make people consider death and the fate of their organs.

32.

In Japan, the incidence of stomach cancer is very high, while that of bowel cancer is very low. But when Japanese people immigrate to Hawaii, this is reversed – the rate of bowel cancer increases but the rate of stomach cancer is reduced in the next generation. All this is related to nutrition – the diets of Japanese in Hawaii are different from those in Japan.

Rachel: "The same diet as in Hawaii should be propagated in Japan also."

Amanda: "Bowel cancer is less severe than stomach cancer."

If the information in the box is true, whose reasoning is correct?

- A Rachel only
- B Amanda only
- C Both Rachel and Amanda
- D Neither Rachel nor Amanda

33. In Country X, car registration plates have two letters followed by two digits.

- The allowed letters are A, B, C, D and E.
- The allowed digits are 1, 2 and 3.
- The two letters must be different from each other.
- The two digits can be the same but must not add up to 4 as this is considered unlucky.

How many different registration plates could there be?

- A 80
- B 120
- C 150
- D 180

34. Shown below are the three small pieces of a puzzle.

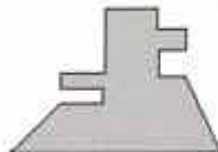


Which one of the following big pieces can be combined with the three small pieces to make a square?

A



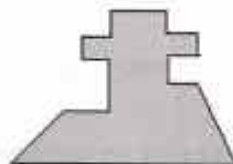
B



C



D



35. Sarah wants to copy some pamphlets to raise awareness for a new charity. She needs 60 copies that are double-sided and in colour. The prices for her local print shop are shown in the table below.

	Black and White	Colour
Photocopying	5¢ (per side)	50¢ (per side)
Printing	10¢ (per side)	80¢ (per side)
Scanning	20¢ (per side)	40¢ (per side)

Assuming that for every \$5, there is a discount of 50¢, what is the price Sarah would need to pay?

- A \$42
B \$86.40
C \$54
D \$60
36. To be able to do an audition for The Performance School, a singer should be able to play at least two instruments. If they could not play two instruments, they either had to be able to sing two octaves or be able to dance well.
- Penelope is an experienced dancer and can play the piano well. However, she could not do an audition for the school.
- What must have been the reason?
- A Penelope was a bad dancer.
B Penelope needed to learn another instrument.
C Penelope should have learnt to sing two octaves.
D Penelope could not sing at all.

37. Whoever cooked the meal must have had both experience and money.

If the information in the box is true, which one of these sentences must also be true?

- A If Clare had both experience and money, she must have cooked the meal.
- B If Clare did not cook the meal, she cannot have had experience.
- C If Clare did not cook the meal, she cannot have had money.
- D If Clare did not have experience, she cannot have cooked the meal.

38. At the end of every baseball season, baseball players at The High School are ranked in order of the number of home runs they got in total over three games. Billy and Joe are baseball players at The High School who played this season. Billy came first place while Joe came fourth place.

Billy: "Since I have a better rank than you then I must have gotten more home runs than you in the third game."

Joe: "And since I have a worse rank than you then I must have hit less home runs than you overall this season."

If the information in the box is true, whose reasoning is correct?

- A Billy only
- B Joe only
- C Both Billy and Joe
- D Neither Billy nor Joe

39. In the Maths Olympiad, there are 5 questions. Whoever gets 80% has a chance to advance to the next stage. However, if you get 60% and answer the last two questions correctly, you can also advance to the next stage. All the students in James' class sat the exam and the same test.

James: "My teacher told me that 6 people in my class got 80% and 9 people got the last two questions correct. That means 15 people will be going to the next stage!"

Which of the following reasons shows the mistake James has made?

- A Some students that got 80% did not do the correct test.
- B The students who got 80% may have also gotten the last two questions correct.
- C Students who got 60% were not mentioned, so the number of students who advanced to the next stage is unknown.
- D We do not know the total number of students in James' class.

40. Using card to pay for items is now just as common as cash. However, using card presents higher risks than cash. If major errors occur because one's bank experiences major issues, one can lose all their money. Furthermore, by using card, experienced trackers can track your locations based on your spending. Additionally, card is highly reliant on technology and electricity. Due to these several reasons, individuals should be using more cash than card.

If the above passage is true, which of the following statement best **weakens** the argument?

- A Being able to track one's location is beneficial when they are faced with issues, such as needing to prove they were at a particular place at a particular time.
- B Cash provides a more visible representation of one's spending habits and may allow people to reconsider spending too much money.
- C Card is a more convenient and faster option compared to cash.
- D Tracking your spending is much easier on card and therefore can allow people to consider saving if they see that they are spending too much.

Test 3

1. Due to a lull in human movement during lockdowns, along with suspended ferry services between Hong Kong and Macau since March, sightings of Hong Kong's endangered pink dolphins have risen by 30 percent.

The local population of Indo-Pacific humpback dolphins reside in murky waters with little sun, and therefore lack pigmentation. Their bubble gum pink appearance is the result of warm blood pumping through vessels near the surface of the skin.

Scientists estimate that only 2,500 of these rare pink dolphins are left in the wild. But research suggests that the vulnerable marine animals have adapted to the more tranquil environment quicker than expected, and could even be making something of a comeback.

Which one of the following conclusions can be drawn from the information provided?

- A The operation of ferry services has created murky water and a comfortable environment for the pink dolphins.
 - B The pink dolphins are rapidly decreasing in number.
 - C Animal populations are disturbed by human activity.
 - D Some species prefer human companionship.
2. A slow train departed from Central Station at 9:30 am and arrived at Newcastle at 1:00 pm. An express train departed from Central Station later and arrived at Newcastle at the same time as the slow train. The express train was three times as fast as the slow train. Each train travelled at a constant speed and did not stop along the way.

When did the express train depart from Central Station?

- A 10:40 am
- B 11:00 am
- C 12:10 am
- D 11:50 am

3. There are five siblings who all play different sports and different instruments. All of them are a different age. We also know that:

- Charlie does track and is one year older than Brian, who is the youngest and plays the trumpet.
- Abby plays the violin.
- Emily plays the drums and does not play basketball.
- The person who can play basketball cannot play the piano.
- David plays volleyball; he is one year older than Emily and two years older than Abby.
- The person who plays violin is the middle child.
- The person who plays the flute also plays volleyball.
- The person who plays soccer is one year older than the person who swims.

What instrument does Charlie play?

- A Drum
- B Flute
- C Piano
- D Trumpet

4. When it rains, the computers sometimes stop working. Computers are known to require certain operating conditions in order to work; however, it cannot be a coincidence that rain outside stops a computer working inside a building.

Which of the following best **weakens** this argument?

- A The computer is sensitive to water.
- B The computer never works.
- C The computer is noted to stop work during all weather conditions.
- D The computer is noted to only work within certain conditions.

5. Encouraging children to aim high and make the most of their capabilities can only serve to increase their sense of self-confidence and improve their attitude towards learning. Children who have long term ambitions to be anything from a mechanic to an actress should be more focused on the tools they need to fulfil this career aspiration. In addition, having a central aspiration may mean that children are more aware of both their strengths and weaknesses. Rather than aiming for mediocre career goals, aspiration helps children give their all.

Which one of these statements, if true, most **weakens** the above argument?

- A Scientists believe that secretaries are happier people than astronauts.
 - B Children who fail to achieve their goals suffer from disappointment.
 - C Students struggle to seek help to overcome their weakness.
 - D Being an actress requires going through many rigorous auditions.
6. Samantha wants to make as many cupcakes as possible from the ingredients she has at home.

The recipe to make 4 cupcakes needs 50 g of flour, 50 g of sugar, 50 g of butter and 1 egg.

At home, she has 300 g of flour, 500 g of sugar, 300 g of butter and 5 eggs.

What is the maximum number of cupcakes Samantha can make?

- A 10
- B 20
- C 24
- D 40

7. Isabella wanted to join the school's dance team for the first time. As they were entering an important competition, the coach wanted to select experienced or talented dancers. Those who have joined the dance team before must send a video of a dance they have done previously. Those who have not joined the team before can learn a dance choreographed by the coach in one week and audition with that dance. They must all pass an interview to show that they are keen participants and are committed learners. If they do not get into the team this time, they can join the school dance team by auditioning after the competition by applying online.

The coach was amazed at Isabella's commitment to the dance team when she talked to her and was astounded by her keen interest. However, she did not get to join the dance team.

What must have been the reason?

- A She did not send a video of a dance that she did before.
 - B She did not apply for a form online.
 - C She did not pass the interviews.
 - D She did not learn the choreography correctly.
8. Each member of a group of 100 students was asked whether they play the piano and whether they play the violin. 70 students said that they play the piano and 35 said that they play the violin.

Which one of the following statements **cannot** be deduced from the information given?

- A At least 5 students play both the piano and violin.
- B At least twice as many students play only the piano as play only the violin.
- C At most 35 students play both the piano and the violin.
- D Every student plays at least one of the piano and the violin.

9. Mr John set a test for his students last week which was made up of three sections. All three sections had 10 questions. Questions in Section A were worth 2 marks each, those in Section B were worth 3 marks each and those in Section C were worth 5 marks each. Mr John has set the pass mark at 70 out of 100.

What is the difference between the largest number of questions that could be correctly answered while failing the test and the smallest number of questions that could be correctly answered while passing the test?

- A 1
- B 3
- C 6
- D 10

10. Sandy is planning a party next month. She invited all her work colleagues to the party, expecting just 25 of them to be able to attend. She calculated that she needed to spend \$8 on food and drink for each of the guests and bought enough to cater for up to 35 guests. When she had received all of the replies she found that 40 of the people she had invited were going to attend. She decided to provide less food and drink, spending only \$7 per guest on food and drink.

How much more did Joanna need to spend to have the amount of food she wanted for the party?

- A \$0
- B \$70
- C \$80
- D \$105

11. Gerry has folded a square piece of paper twice, creating a smaller square, and cut through the resulting four layers to produce the following appearance:



Which one of the following could **not** be the paper once Gerry has unfolded it?

A



B



C



D



12. The price for passes for a theme park are shown in the table below.

Pass for 1 day	Pass for 4 consecutive days	Pass for 7 consecutive days
\$35	\$70	\$140

I will need a pass on the following days of the month: 1, 2, 3, 5, 10, 11, 12, 14, 15, 16.

How much is the cheapest valid combination I could use?

- A \$175
B \$210
C \$245
D \$280

13. I have a basket containing three types of fruits. All but four are oranges, all but four are apples and all but four are bananas.

How many pieces of fruit are in the basket?

- A 6
- B 8
- C 10
- D 12

14. In a music class, there are five students. Throughout the year, they each take three tests, each marked out of 100. The results of the tests, as well as each student's total score are shown in the table below. At the end of the year, a prize is awarded to the student with the highest total score. Unfortunately, due to a printing error, some of the scores were lost (shown in cells with a black fill).

Student	Test 1	Test 2	Test 3	Total Score
Nell	64	46	60	
Zane	80	19		138
Cedrick	47		15	101
Edward	31	90		167
Cindy	49	72	57	

From the numbers still visible, which student will get the prize?

- A Nell
- B Cindy
- C Cedrick
- D Edward

15. If Christine starts dance classes, she can participate in the talent show.

If she participates in the talent show, she can win a trophy.

If she does not win the trophy, then she may win a medal instead. No other awards are given.

If the above statements are correct, which one of the following is not possible?

- A Christine starts dance classes but does not participate in the talent show.
 - B Christine starts dance classes but never receives a medal.
 - C Christine wins both a trophy and a medal at the talent show.
 - D Christine participates in a talent show and wins a medal.
16. Sally's school has a policy that students must hand in permission notes to go on school excursions before the deadline or else the teacher will be upset. If the teacher is upset, they will likely forbid the student from going on the excursion.

Sally: "Well, the teacher was upset with me and forbid me from going to the excursion. This means that I did not hand in the permission note on time."

Which one of the following sentences shows the mistake Sally has made?

- A Just because Sally handed in the note late, it does not mean that the teacher will be upset.
- B Just because Sally handed in the note late, it does not mean that she was forbidden from going on the excursion.
- C Just because the teacher was upset with Sally, it does not mean that the teacher was not upset with other students.
- D Just because Sally was not allowed on the excursion, it does not mean that she handed in the note late.

17. For a children's holiday activity, all groups must have a minimum ratio of adults to children of 1 : 4. The ratio of boys to girls must be either 1 : 2 or 2 : 1. Four groups are currently composed as follows.

Which group meets the ratio for the holiday activity?

Group	Number of adults	Number of boys	Number of girls
A	3	6	4
B	4	8	8
C	5	15	5
D	6	16	8

18. Lydia believes that many teens are less likely to engage in activities like the older generations did, such as going to see friends and outdoor activities. There are worrying signs that teens' use of social media networks like Facebook and Snapchat is contributing to a mental health crisis in their age group. Phones also seem to be getting in the way of sleep, which has been directly linked to later emotional issues in teens.

Which one of these statements, if true, most **strengthens** Lydia's argument?

- A Social media has allowed teens to make more friends.
- B Using phones allows people to become more organised.
- C Social interactions can improve mental health.
- D Not all teens have access to phones.

19. Tom, Sam and Ken are to complete the tasks in the following table. Each task can be completed by anyone, and any task can be done at the same time as each other. However, only one person can work on a task. Furthermore, to be fair, each person should work for roughly the same amount of time.

Task No.	Task	Time Needed (Minutes)
1	Wash the car	12
2	Clean the dishes	9
3	Mow the lawn	15
4	Make the bed	4
5	Pat the cat	2
6	Feed the dog	5
7	Plant the flowers	15
8	Vacuum the house	22

What is the shortest time in which they can complete all the tasks?

- A 26 minutes
- B 27 minutes
- C 28 minutes
- D 29 minutes

20. Josephine is doing some online shopping and she is trying different coupon codes to get a discount on her purchase. There has been a list of possible coupon codes published online that she needs to decipher.

She knows that A = 1, B = 2 up to I = 9 and J = 0.

She has been told that codes which start with even numbers do not work.

The following codes are on the list:

D F H A

C G H B

B E C G

H G D A

What is the coupon code for this website?

A 1 3 2 5

B 3 7 8 2

C 5 2 6 1

D 9 4 2 6

21. There are five oranges and five apples in a basket. Emily and Sarah evenly share the fruits in the basket. They both got five fruits each.

Emily: "If I got more oranges than you then you must have more apples than me."

Sarah: "And if I got 1 less orange than you then you must have 1 less apple than me."

If the information provided is true, whose reasoning is correct?

A Emily only

B Sarah only

C Both Emily and Sarah

D Neither Emily nor Sarah

22. Adults thinking back on their experience of school almost always remember with most gratitude the teachers who taught them more than just a subject, and prepared them for life, not just for an exam. But the current generation of students may not have the same experiences to look back on in future years because the role of teachers has changed. The focus for both students and teachers is now mainly on exam results, and that is how their success or failure is measured. So, children today receive a less valuable education than their parents and grandparents did.

Which of these statements, if true, most **strengthens** the above argument?

- A Education for life is of greater value than passing exams.
 - B Adults looking back on their childhood often remember only the good parts.
 - C Many young people have higher ambitions than becoming like their teachers.
 - D Exam results are the vital keys which open the door to education and a livelihood.
23. Why do so many people want to live in a house on a street? Living in an apartment is much better. First, an apartment is most likely smaller than a house and thus, maintenance of the property will be much easier. Second, in an apartment, you have neighbours surrounding you. If you are in an emergency, as long as you make some noise, your neighbours are likely to hear you and help you out. And finally, living in an apartment can provide you with a beautiful view of the city from the balcony. Therefore, living in an apartment is much better than living in a house.

If the above passage is true, which of the following statement best **weakens** the argument?

- A Only high-rise apartments near the beaches have nice views.
- B Living in a high-rise apartment makes escaping from it quite difficult in the case of a fire.
- C Living in high-rise apartments is not suitable for those with a fear of heights.
- D Falling from a high-rise apartment is highly likely fatal.

24. A farm is looking to purchase 1000 bales of hay for their cows and have found three suppliers.

Which is the most economical option for them?

Supplier	Cost (\$) (Per bale)	Discount
Koles	35	10% off for orders exceeding 500 bales
Woolly	25	5% off orders exceeding \$1000
Baldy	30	Buy 9 bales, get the 10 th free
Lohay	40	Buy 4 get and one free

- A Koles
 B Woolly
 C Baldy
 D Multiple suppliers have the same price
25. When Jessica's car is unlocked when parked, her phone will receive an alert. Otherwise, nothing happens to the phone.

Jessica: "I received an alert on my phone and went to lock the car. But I'm still getting the alert on my phone, so that means the car has not registered that it is locked."

Whitney: "No, that's not right. The alert on the phone means that the car is still unlocked."

If the information in the box is true, whose reasoning is correct?

- A Jessica only
 B Whitney only
 C Both Jessica and Whitney
 D Neither Jessica nor Whitney

26. A ferryboat is full when it has ten cars on board. It is also full when it has six trucks on board. The ferryboat never carries cars and trucks at the same time. The ferryboat made seven trips across the river and was full on each trip. It carried a total of 54 vehicles (cars and trucks) across the river.

How many trucks did the ferryboat carry altogether in the seven trips?

- A 24
- B 28
- C 30
- D 36

27. The sales figures for five different brands of coffee machines are given in the table below, collected over a one-week period.

Brand	Sales
1	381
2	302
3	155
4	387
5	392

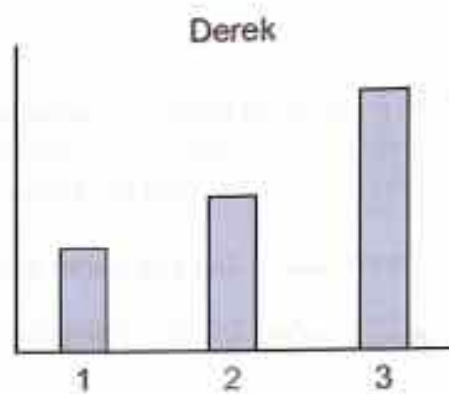
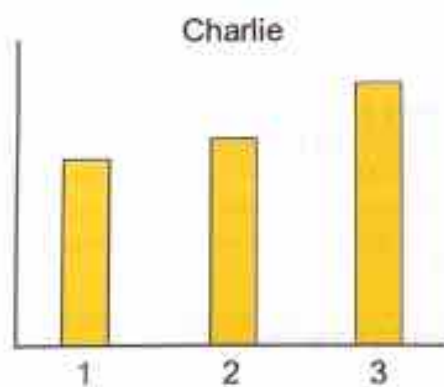
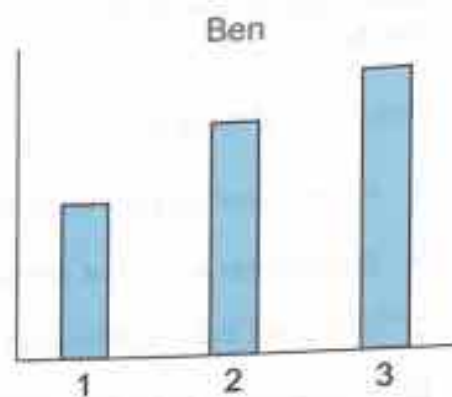
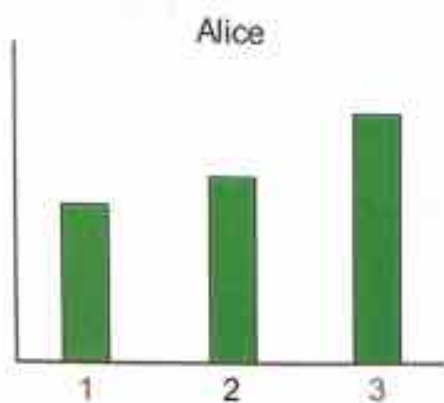


The data was then taken and transformed into a graph; however, one brand of coffee machine was accidentally omitted. Which brand was omitted?

- A Brand 3
- B Brand 2
- C Brand 5
- D Brand 1

28. A company gives a bonus to the salesperson who had the greatest increase in sales from Year 2 to Year 3. Four sales people's sales are shown in the graphs below.

Which employee received the bonus?



- A Alice
- B Ben
- C Charlie
- D Derek

29. Vegans tend to be thinner and have a lower body mass index (BMI) than non-vegans. This might explain why an increasing number of people turn to vegan diets as a way to lose excess weight. Adopting a vegan diet may help keep your blood sugar in check and Type 2 diabetes at bay. Several studies show that vegans benefit from lower blood sugar levels, higher insulin sensitivity and up to a 78% lower risk of developing Type 2 diabetes than non-vegans. From this, it is clear to see just how beneficial a vegan diet can be.

Which one of these statements, if true, most **weakens** the above argument?

- A A vegan diet may not taste as good as a non-vegan diet.
 - B Blood sugar levels that are too low can be fatal.
 - C BMI is not the only indication of health.
 - D Insulin needs to be injected into the bodies of those with diabetes.
30. In a survey of teachers, everyone who liked teaching liked preparing for lessons. Everyone who liked preparing for lessons liked cleaning up, but no one who liked preparing for lessons liked organising.

David, Clare, Jim and Sarah all took part in the survey.

Based on the above information, which one of the following must be true?

- A If David likes cleaning up, he also likes preparing for lessons.
- B If Clare does not like organising, she does not like cleaning up.
- C If Sarah likes teaching, she does not like organising.
- D If Jim does not like teaching, he does not like preparing for lessons.

31. Cameron was starting at a new school. His friend Jack told Cameron to say hi to Harry, who Jack knew from soccer. Jack said Harry had freckles and red hair. At lunch time, Cameron went to the soccer field.

Cameron: I can only see one person here that has red hair and freckles. He must be Harry.

Which one of the following sentences shows the mistake Cameron has made?

- A There were both boys and girls playing on the fields.
 - B There were some kids playing rugby, and some playing soccer.
 - C Cameron didn't know whether Harry was tall or short, but the boy Cameron saw was short.
 - D Harry might not be playing soccer today.
32. Eating chocolate for breakfast is not a bad idea at all. Chocolate is a treat which many people love. Many young people struggle to get out of bed in the morning and so the thought of a delicious chocolate bar could be enough to tempt them from their beds and therefore get to school on time.

Which one of the following statements, if true, most **weakens** the above argument?

- A Having more than the recommended daily intakes of chocolate can cause obesity in children.
- B Eating chocolate causes people to be sleepy in the middle of the day.
- C Young children cannot wake up without the temptation of food.
- D Chocolate is too expensive in large quantities.

33. Tryouts for the school's baseball team happen every year. Each player needs to pass the tryouts to get into the team.

Darren: "I passed the tryouts and am currently in the school's baseball team. This means I will definitely be on the team next year."

Which one of the following sentences shows the mistake Darren has made?

- A If Darren was good enough to pass this year, he will be good enough to pass next year.
 - B The school's baseball team has a limited number of spots.
 - C Darren had a fight with one of teammates this year.
 - D There will be separate tryouts for next year's team.
34. Some scientists were measuring the heights of different trees in a local forest. This was what was found:
- The shortest tree was a spruce tree.
 - The tallest tree was an oak tree.
 - Pine trees range from three to five metres tall.

Which one of these sentences can be concluded from the above information?

- A All oak trees are taller than all spruce trees.
- B At least one oak tree is more than five metres tall.
- C All spruce trees are shorter than all oak trees.
- D The shortest tree is four metres tall.

35. A word is made into a code using the following rule.

The first letter remains the same.

The second letter is changed to the letter that is one letter before that.

The third letter is changed to the letter that is two letters before that, and so on.

For example, SOLEMN is changed to SNJBII.

In the same code, what is represented by JNWLQN?

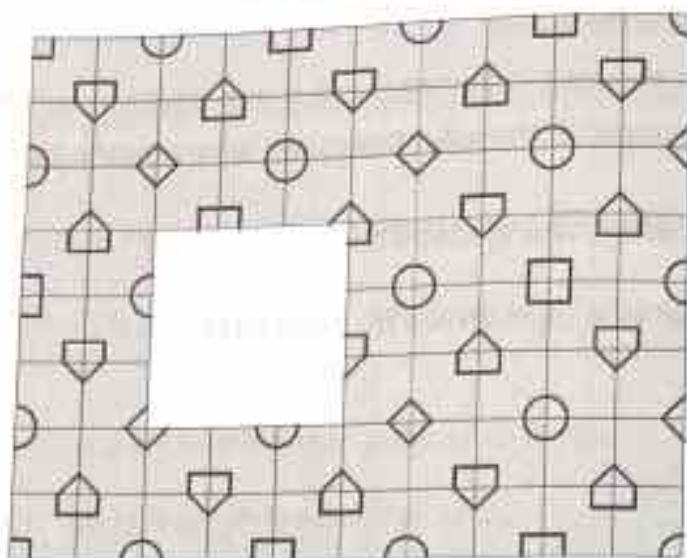
A JAGGED

B JOVIAL

C JOYFUL

D JOYOUS

36.



This is part of a tiled wall in my house that has a regular, repeating pattern. Nine tiles have fallen off the wall and broken, leaving a square space.

Which one of the following tiles could be used to replace the central tile of the group of nine?

A



B



C



D



37. Monthly new car sales during 2019 are shown in the table below.

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
154	68	464	176	194	228	172	72	425	157	172	166

Steve is asked to produce a chart showing this information.

Which of the following is the chart Steve should produce?

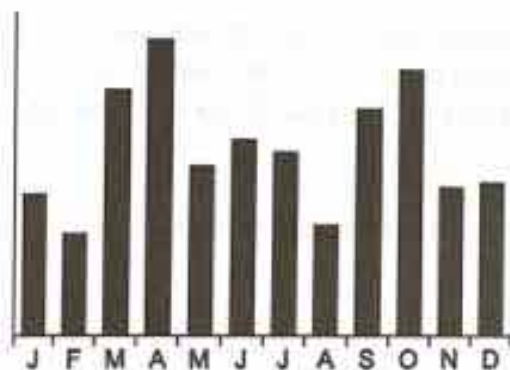
A



B



C



D



38. A rocket company, BOOM makes the following claim about their new booster, "Our new TwinTurbo Booster is, on average, twice as efficient as our competitors."
- The following data was collected; however, some omissions have been made.

Company	Trial 1	Trial 2	Average
	All values are /150		
BOOM	107		
Competitor 1	100	99	99.5
Competitor 2	108	87	
Competitor 3			118.5

How many values need to be retested and found in order to successfully test the claim?

- A 0
- B 1
- C 2
- D 3
39. Ms Mason made her art class complete two assignments for the end of year exams. One was a theory exam and the other was a practical exam. These two scores were then added together to give the student's overall score for the year. Emily and Sarah got different overall scores this year.
- Emily:** "If I got 10 points less than you in the theory exam, then you must have gotten 10 more points than me in the practical exam."
- Sarah:** "And if you got a higher score than me in the theory exam, then I must have gotten a lower score than you in the practical exam."

If the information above is true, whose reasoning is correct?

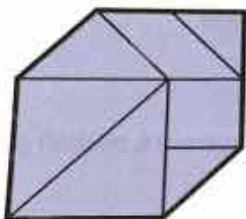
- A Emily only
- B Sarah only
- C Neither Emily nor Sarah
- D Both Emily and Sarah

40. The tangram, known as a Chinese puzzle, is a collection of seven polygons, called tans: five isosceles right triangles (two large, one medium and two small triangles), a square with sides of length 1 and a parallelogram.

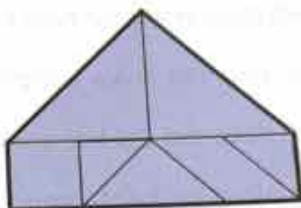
These seven pieces are arranged.

Which one of the following is not a tangram?

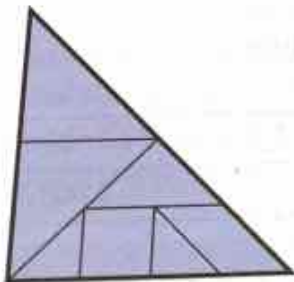
A



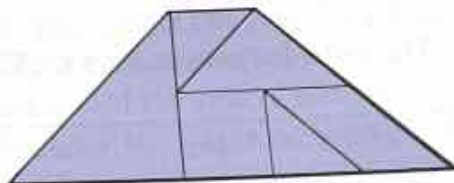
B



C



D



1. If Susan wears high heel shoes, she will roll her ankle.

If she rolls her ankle, she will need an ice pack if there is swelling present.

If she does not use an ice pack to control the swelling, she cannot play at tomorrow's netball game.

If the above statements are correct, which one of the following is not possible?

- A Susan does not wear high heel shoes and rolls her ankle.
 - B Susan rolls her ankle but does not need an ice pack.
 - C Susan does not roll her ankle but does not play at tomorrow's netball game.
 - D Susan plays netball tomorrow with a swollen ankle.
2. The price for passes for a ski lift are shown in the table below.

Pass for 1 day	Pass for 3 consecutive days	Pass for 7 consecutive days
\$35	\$70	\$140

I will need a pass on the following days of the month: 1, 2, 3, 5, 6, 10, 11, 12, 14, 15, 16.

How much is the cheapest valid combination of ski lift passes I could use?

- A \$175
- B \$240
- C \$245
- D \$280

3. Elise broke her necklace.

She found one quarter of the beads on the carpet, one third down the side of the stool, one fifth in her bag, while one sixth remained in her hand. Five beads were never found.

How many beads were there to start with?

- A 100
B 74
C 56
D 30
4. In a mathematics tutoring class, there are five students. Throughout the year, they each take three tests, each marked out of 100. The results of the tests, as well as each student's total score are shown in the table below. At the end of the year, a prize is awarded to the student with the highest total score. Unfortunately, due to a printing error, some of the scores were lost this year.

Student	Test 1	Test 2	Test 3	Total Score
Nick	64	18	70	
Jack	80	19		138
Chris	47		15	101
Ethan	31	90		167
Cindy	49	72	57	

How many numbers need to be found to determine which student will get the prize?

- A 0
B 1
C 2
D 3

5. James is going on a camping trip. At the camping site, there is a facility that sells different kinds of equipment. There is a notice board with prices of equipment listed:

Equipment	Price (\$)	Number left in stock
Gas	5	5
Tent	15	2
Pan	2	2

James currently has \$30. He wants to buy as much equipment as possible and spend his entire budget. With this goal in mind, how many pieces of equipment can he buy?

- A 2
- B 3
- C 4
- D 5

6. There are fifty positions in the school musical. Terry's teacher has stated that those who can dance have a higher chance of getting a role in the musical.

Terry: "Well, I can't dance at all. This means I don't have a chance of getting a role in the musical when I audition."

Jake: "No, that's not true. There are only two professional dancers in the school, so you still have a very high chance."

If the information provided is true, whose reasoning is correct?

- A Terry only
- B Jake only
- C Both Terry and Jake
- D Neither Terry nor Jake

7. It is reported that although Vitamin E, present in fresh fruits and vegetables is beneficial for the human body, capsule Vitamin E does not have the same effect.

Which one of these statements, if true, most **strengthens** the argument in the box?

- A Vitamin E is crucial to the health of animals.
 - B Vitamin E in capsules allows healthier hair to grow.
 - C Only Vitamin E in fruits is absorbed entirely by the small intestine.
 - D The sale of capsule Vitamin E should be banned.
8. Max, Alex and Kevin are playing basketball. They take turns to shoot the basketball into the hoop.
- If the ball goes through the hoop, then that player must receive \$5 from each of the two other players.
 - If the ball misses the hoop, then that player must give \$5 to each of the other two players.

Only Max and Alex's shots go through the hoop.
If Max started with \$20, how much does he have now?

- A \$15
- B \$20
- C \$25
- D \$30

9. In a group of hockey players, those who played tennis played soccer. Those who played soccer played baseball, but no one who played soccer played cricket.

David, Aaron, Lia and John are all in the group.

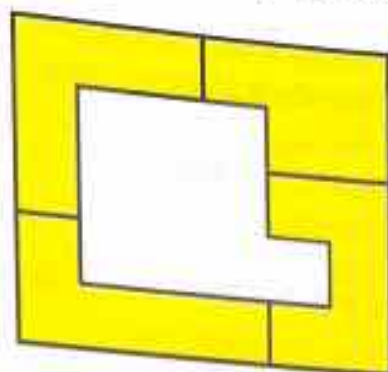
Based on the above information, which one of the following must be true?

- A There are four different sports the players can like.
 - B Aaron, a tennis player, plays at least three sports.
 - C If Lia plays baseball, she also plays cricket.
 - D If John plays soccer, he must like baseball.
10. Every year there are nearly 25 000 car fires, but not all drivers travel with a fire extinguisher in their car. Sam thinks that if more drivers could be encouraged to carry fire extinguishers then the number of car fires could be considerably reduced.

Which one of these statements, if true, most **weakens** Sam's argument?

- A Research shows that fire extinguishers are very effective in most car fires.
- B Most drivers never experience a car fire.
- C There is a possibility that the car fires cannot be put out with an extinguisher.
- D Whether a car is carrying an extinguisher or not does not affect the possibility of a car fire.

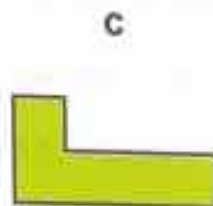
11. A pentomino is a tile made up of 5 squares which are joined along their edges. Four pentominoes have been placed together in the pattern shown below.



Two more pentominoes are to be placed to fill the gap. One of them is a new shape, while the other is a repeat of one of the pentominoes already positioned. The new shape is shown below.



Which is the other pentomino that is to be placed in the gap?



12. Lucy lives in Newtown. On Saturday she will visit her friend Martha who lives in Midtown, and she will arrive in time for lunch at 12:30.

Her journey will consist of a 10-minute walk to Newtown station, then a train ride to Midtown station and finally a bus ride to Martha's house.

Trains leave Newtown at 06:45 and then every 30 minutes until 21:15, taking 1 hour 20 minutes to reach Midtown.

Buses leave Midtown at 08:00 and then every 20 minutes until 21:00, arriving at Martha's house 12 minutes later.

What is the latest time that Lucy can leave home so that she arrives at Martha's house in time for lunch?

- A 10:05
- B 10:15
- C 10:35
- D 10:45

13. I saw 9 cars with bicycles on the roof in the parking lot.

Some of the cars were carrying one bicycle, and some had two.

Counting the wheels of the cars and the bicycles, there was a total of 60 wheels.

How many cars had 2 bicycles on the roof?

- A 1
- B 2
- C 3
- D 4

14. Five athletes participate in a race where there are six hurdles, spaced along a 100 m distance. They aim to finish the race in the shortest amount of time and with the least number of hurdles knocked over.

- Harry was faster than Gina, and knocked over three hurdles.
- Nathan finished second, and knocked over two hurdles.
- Aiden did not knock any hurdles over, and took the longest.
- Kenny wasn't as fast as Nathan, and Nathan was not as fast as Harry.
- Gina knocked over fewer hurdles than Nathan, and finished after him.

If all the above statements are true, only one of the sentences below cannot be true. Which one?

- A Gina knocked over more hurdles than Aiden.
 - B Harry was faster than Aiden.
 - C Harry knocked over the most hurdles.
 - D Harry was not first to finish.
15. Miss Brown knows that exactly one of her four students stole a mint from the candy jar. She knows that if the student was late to class and was too full to eat at recess, then they must have stolen the candy.

<i>student</i>	<i>Were they late to class?</i>	<i>Were they full at recess?</i>
1	(a)	Yes
2	Yes	(c)
3	(b)	No
4	No	(d)

Which two of the missing answers must be known in order to determine which student stole a mint from the candy jar?

- A (a) and (c)
- B (b) and (d)
- C (b) and (c)
- D (c) and (d)

16. Victor is visiting the Green Coast and is at the theme park. He visits the following rides:

Hair-Raiser*, Cow-a-bunga*, CalClock, O-Matic 1000*, Coney Drop, Ghost Train*,
Double Dragon of Fire*, Barrels of Fun*, Power Surge and Hair Raise.

Pricing is given in the table below:

1 Ride	Day Pass	
\$10	\$90	\$120 for rides including * or *
* Rides cost an extra \$5		
* Rides cost an extra \$10		

How much does Victor save by buying the Day Pass instead of buying the individual tickets?

- A \$30
B \$10
C \$11
D \$20

17. The winner of the annual pie baking contest is selected using a few criteria. If what they baked is a fruit pie, it must be filled with various types of fruits. If it is a savoury pie, it has more chance of winning but must contain minced beef. If the pie does not contain fruits or meat, it can contain custard.

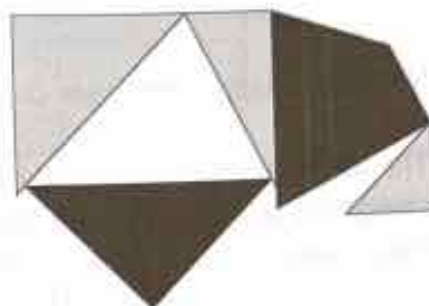
George enters his apple pie into the contest, but it was not considered to be a possible winner in the contest.

What must have been the reason?

- A There were too many other sweet pies in the contest.
B It had more chance of winning if it contained custard.
C He should have added extra ingredients.
D He should have tried making a savoury pie instead.

18. Kevin is going to make some designs using two identical square tiles that are black on one side and grey on the other. He will cut the tiles and rearrange the pieces, without discarding any.

Which of the following designs is not one that Kevin could make?

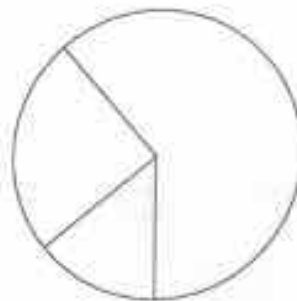
A**B****C****D**

19. 250 people were asked to name their favourite holiday destination.

King Island	43
Silver Coast	22
Mac Port	77
Kangaroo Resort	108

No other destinations were chosen.

The information was entered into a computer in order to construct a pie chart. However, one of the destinations was left out. As a result, this was the pie chart that appeared on the screen.



Which destination was left out?

- A King Island
- B Silver Coast
- C Mac Port
- D Kangaroo Resort

20. Peter is planning a picnic and has prepared a list of items that he wants to purchase. He has mixed up the data on his list but is certain that the strawberries are listed next to the correct amount in the table below.

Item	Cost (\$)
Strawberry	17
Crisps	32
Biscuits	21
Drinks	15

Peter is sure that the combined amount for the strawberry and the crisps was less than the combined amount for the biscuits and drinks. He is also sure that the combined amount for the biscuits and crisps exceeded the combined amount for the strawberries and the drinks.

Which of the following statements cannot be correct?

- A The biscuits cost more than the drinks.
 - B The drinks cost more than the strawberries.
 - C The strawberries cost more than the biscuits.
 - D The crisps were the least expensive item on the list.
21. It's true that laughter is a strong medicine. It draws people together in ways that trigger healthy emotional changes in the body. Laughter boosts the mood which protects you from the damaging effects of stress. Nothing works faster or more dependably to bring your mind and body back into balance than a good laugh. Humour lightens your burdens, inspires hope, connects you to others, and keeps you grounded, focused, and alert. It also helps you release anger and forgive sooner.

Which one of the following, if true, most **strengthens** the above argument?

- A Laughing strengthens an individual's core muscles.
- B People who laugh are more likely to be loud in public.
- C People who laugh are able to overcome struggles in the workplace.
- D Laughter allows a person to externalise their happiness.

22. The number of ice cream and candy sales made from a particular convenience store was recorded from Monday to Friday.

- On Thursday, the second highest amount of ice creams were sold, but only two candies were sold.
- Tuesday had more candy sales than Wednesday, but not as many as Friday.
- Monday had more ice cream sales and more candy sales than Friday.
- Tuesday broke the record for most ice cream sales in a day.

If all the above statements are true, only one of the sentences below cannot be true.

Which one?

- A Wednesday had fewer ice cream sales than Tuesday.
- B Monday had the most candy sales.
- C Friday had the third most ice cream sales.
- D Friday had the second most candy sales.

23. As part of the process of hiring new staff, all applicants are required to take two tests. Test 1 has a maximum score of 20, whilst Test 2 has a maximum score of 25. Applicants are only told their total score.

Any applicant who scores at least 15 on Test 1 moves onto the next stage automatically. Of the remaining applicants, those who score at least 20 on Test 2 move to the next stage.

When Holly was told her total score, she was certain that she would be moving onto the next stage of the process.

What is the smallest possible total score she could have been told?

- A 33
- B 34
- C 35
- D 36

24. There are 5 students in an English tutoring class. Throughout the year, they each took three tests, with the average then calculated for each student. The student with the highest average gets a prize.

Whilst the results were being read out to the class, the teacher accidentally spilt some coffee, resulting in some of the data being lost.

Student	Test 1 (%)	Test 2 (%)	Test 3 (%)	Average (nearest %)
1		67	24	47
2	29		98	54
3	26			62
4	41	31	90	
5			54	56

*Cells in grey represent lost data.

From the results still legible, which student will get the prize?

- A Student 3
- B Student 4
- C Student 2
- D Student 5

25. Matt writes and records his own songs. He has been asked to produce a list of his songs. The album is to be played throughout the course of an event. The event will last for 3 hours and each song will be played exactly three times. Each of Matt's songs lasts at least 2.5 minutes and at most 4 minutes.

What is the minimum number of different songs that Matt can include so that all the conditions are met?

- A 10
- B 15
- C 16
- D 24

26. Liam, Mary and Ben have 30 \$1 coins each.

They take turns to roll a dice.

- If the number rolled is 1, 2 or 3, then that player gives the number of coins shown on the dice to each of the other two players.
- If the number rolled is 4, 5 or 6, then that player receives the number of coins shown on the dice from each of the other two players.

Liam rolls a 1.

Mary rolls a 3.

If the friends end up with \$27, \$21 and \$42 respectively, which number did Ben roll?

- A 2
- B 3
- C 4
- D 5

27. Eva attends a very prestigious sports school. Everyone at the school who is good at running also happens to be able to play tennis.

Eva: "I am a professional tennis player so I should also be good at running."

Which one of the following sentences shows the mistake Eva has made?

- A People who are good at tennis have more practice in jogging.
- B People who are good at running can play, but are not excellent tennis players.
- C People who attend the prestigious school are only admitted for one sport.
- D People who can play tennis may not be good at running.

28. There is a statement written on each chest, but only one of the four statements is true.

Chest 1	Chest 2	Chest 3	Chest 4
The gold is in Chest 4 only.	There are 2 chests with gold.	The gold is in every chest except for Chest 4.	The gold is here.

Which chest must have the true statement?

- A Chest 1
 - B Chest 2
 - C Chest 3
 - D Chest 4
29. Tilly, Charlie, Sylvia, Nick and Peter play hopscotch on a grid with the numbers 1-10.
- Only Tilly and Charlie throw their stones onto even numbers.
- Nick's stone lands on a lower number than anyone else's.
- Sylvia's stone lands on 7.
- No two people throw their stone on the same number.
- Nobody's stone lands on the numbers 1, 5 or 10.
- If these statements are true, which one of the sentences below **must** be true?
- A Tilly's stone lands on number 5.
 - B Peter's stone lands on number 9.
 - C Sylvia's stone lands on a higher number than Charlie's.
 - D Nick's stone lands on number 2.

30. Kate wants to buy 6 board games. There are 4 shops that sell the board games that she wants.

The shops, the prices of the board games and any special offers are shown in the following table.

	Standard price of a board game	Special offer
Shop A	\$32	Buy 1, get 2 nd for half price
Shop B	\$33	Buy 2, get the 3 rd one free
Shop C	\$30	Buy 5 or more, get 15% off
Shop D	\$24	none

Kate wants to buy all her board games from the same shop and to spend as little as possible.

Which shop should Kate buy her board games from?

- A Shop A
- B Shop B
- C Shop C
- D Shop D

31. A company packs cups of instant noodles in three different sized boxes that can hold 7, 42 and 55 cups.

Their packing machine releases noodles one at a time or six at a time as it has two settings, with the aim of being as efficient as possible.

If I want to use an equal number of boxes of each size, what settings of the machine should be used to pack the instant noodles?

- A 3×1 at a time for every 17×6 at a time
- B 4×1 at a time for every 14×6 at a time
- C 2×1 at a time for every 17×6 at a time
- D 2×1 at a time for every 14×6 at a time

32 Five people are standing on the same row during the school assembly. Each person owns a different pet and they all have different types of hobby. We also know that:

- Only 2 people own a pet with four legs and they are directly next to each other.
- Bob owns a spider and he does not like to watch movies.
- Emma is standing next to Ashley, who dislikes snakes.
- The person who likes swimming is next to the person who likes reading.
- Dennis likes to take pictures and is standing next to a person who swims.
- Ashley owns a dog and there are 2 people in between her and Dennis.
- Coleman is standing in the middle and he owns a goldfish.
- The person who likes computer games is standing on the right end of the row and owns a cat.

Using the information above, which pet does a swimmer own?

- A Cat
- B Snake
- C Dog
- D Spider

33 In Joanne's class, whoever completes the most homework becomes the student of the month. No one has ever been student of the month twice before.

Joanne: "Well, I was just awarded student of the month last month. This means I won't be able to get the award again, so I should stop doing my homework."

Which one of the following sentences shows the mistake Joanne has made?

- A Joanne is a hardworking student, so she actually has a higher chance of getting the award again.
- B Students in the past did not get the award again because they stopped trying.
- C Joanne can still get the award if she completes the most homework.
- D No one has ever been student of the month before because doing homework is difficult.

34. Jasmine's method of constructing computer passwords is to take a word of the required length and replace some or all of the letters in the word with numbers or symbols that she thinks resemble the letter. She is careful to distinguish between upper and lower case letters.

The only letters she replaces are:

a	b	B	E	g	o	s	S	T	u	x	z
@	6	8	f	9	0	5	\$	7	"	%	2

Two of the passwords she could make from Jasmine are Ja5mine and J@5mine. There is a total of 8 different passwords she can make using her name.

How many different passwords can she construct from the word Schools?

- A 5
B 16
C 32
D 12

35.



The figure above shows four villas in a building. In this building, each villa is occupied by only one person. Alice lives next to Sam, and Paul lives next to Alice and Dara.

In which apartment could Alice live?

- A 2 only
B 3 only
C 2 or 3
D 1 or 4

36. In an Internet-driven world, social media has become the go-to source of all kinds of information. This is especially relevant in crisis-like situations, when warnings and risk-related information are actively circulated on social media. But currently, there is no way of determining the accuracy of the information. This has occasionally resulted in the spread of misinformation, with some readers often bearing the brunt.

Which one of the following, if true, most **strengthens** the above argument?

- A The social media platform Facebook allows users to post information at very fast speeds.
- B The social media platform Twitter has no program to analyse what content is posted.
- C Many politicians use social media as a way to reach out to audiences about their policies.
- D Social media is a place where discussion can occur between any two random people.

37. Adam, Beckie, Cynthia and Douglas are making pancakes according to the recipe below.

To make 12 pancakes you need:

100 g flour

2 eggs

300 ml milk

This table shows the amount of each ingredient that each person has.

	Adam	Beckie	Cynthia	Douglas
Flour (g)	175	150	150	175
Eggs	4	3	4	3
Milk (ml)	500	400	600	550

Which person can make the most pancakes?

- A Adam
- B Beckie
- C Cynthia
- D Douglas

38. A group of friends plans to make a round trip using a boat service along a river, so that they can see the city from the waterfront. The sightseeing journey will start from Chelsea Harbour and take the boat service to both ends, finishing back at Chelsea Harbour. The friends are ready to take their first boat at 07:50. They want as little time as possible waiting for a boat, so they will always get the next available boat.

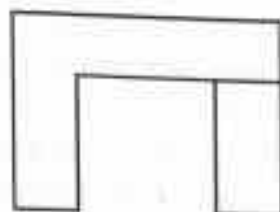
Eastbound					
Putney	07:25	08:00	08:40	09:25	10:10
Wandsworth	07:35	08:10	08:50	09:35	10:20
Chelsea Harbour	07:45	08:20	09:00	09:45	10:30
Cadogan	07:50	08:25	09:05	09:50	10:35
Embankment	08:10	08:45	09:25	10:10	10:55
Blackfriars	08:20	08:55	09:35	10:20	10:05

Westbound				
Blackfriars	07:20	08:25	09:10	09:50
Embankment	07:30	08:35	09:20	10:00
Cadogan	07:50	08:55	09:40	10:20
Chelsea Harbour	07:55	09:00	09:45	10:25
Wandsworth	08:05	09:10	09:55	10:35
Putney	08:15	09:20	10:05	10:45

At what time will they return to Chelsea Harbour?

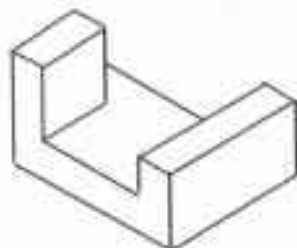
- A 09:40
- B 09:45
- C 10:25
- D 10:30

39. When viewed from one side, an object looks like the picture below.

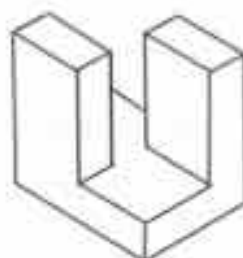


Which one of the following objects could give the views above?

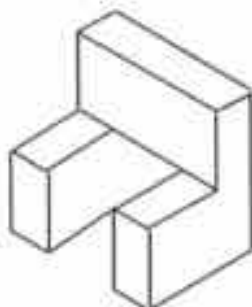
A



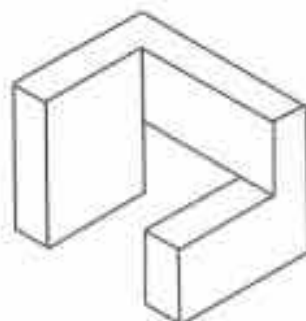
B



C



D



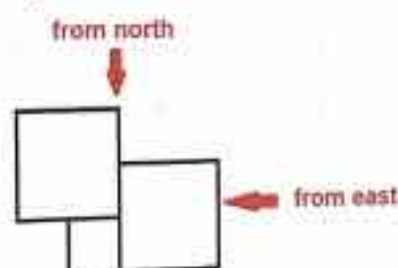
40. As electric cars become more and more popular, environmentalists are hailing the change, welcoming it with open arms. However, they are not a perfect solution, and there are still a lot of problems; problems that can be solved with a simple bicycle. Electric cars are actually more expensive than their petrol-powered counterparts. Getting one can be a costly investment. And then there's bikes. The average cost of a bicycle is a measly \$233. That amount doesn't even touch how much you'd spend in a year on petrol, and it's still a lot less than what you'd spend on charging your electric car.

Which one of these statements, if true, most **weakens** the above argument?

- A Money spent on petrol per year is quite significant for some people.
- B Petrol is a finite resource that is going to run out some day.
- C Bicycles require a lot more energy to get to places.
- D Electric cars are not as powerful as petrol-powered cars.

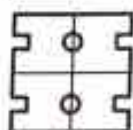
TEST 1

1. **B:** Harry will only buy expensive ice cream if:
 - it is extremely tasty and of average popularity; OR
 - it is of average taste and extremely popular.
 If he did not buy an expensive ice cream, it must not have fulfilled one or more of his requirements.
2. **B:** If a person likes movies, that person likes animations, too. However, this does not mean that if that person likes animations, they must like movies. It is not the same if it goes the other way around.
 If Amanda likes movies, she likes animations. If she likes animations, she likes cartoons but does not like reality shows.
3. **A:** From the north, all three cubes are equally visible. From the east, you can see the cubes stacked vertically with one cube partially visible on the right. Only figure A fits this description.



4. **D:** There is not enough information on other workers in the factory to make a conclusion about them.
5. **C:** The students who used hand sanitiser were the ones who had not been sick.
6. **D:** Paul – Wed → Sat → Mon → Wed → Sat
 Jimmy – Wed → Fri → Sun → Tue → Thu → Sat
 Neil – Wed, next Sat
 The next day they will meet up again is 10 days later, on Saturday.
7. **D:** Lee 8, Sam and Ethan 9, Susan 10, Kerry 11
8. **B:** $30 - 10 = 20$, $100 - 30 - 10 = 60$, $100 + 10 - 30 = 80$
9. **D:** Total number of days absent by 8 people = $40 + 20 + 9 + 8 + 5 = 82$
 Harry = $82 - \text{total days absent by 7 people} = 82 - 66 = 16$ days
10. **C:** From Room 2, if the statement 'I am not in this room' is False, then the statement 'I am in Room 1 or 3' must be False as well; and vice versa. However, this cannot happen as there can be no more than one false statement on each door.
 Therefore, both statements on the door of Room 2 must be true.
11. **B:** $5 \times 30c + 8 \times 50c = \5.50

12. D:



13. A: Charlie – football, Abby – swimming, Brian – soccer
 14. B: Computer screens emit harmful blue lights, which affects the development of young children.

15. D: Lance was paid a bonus, meaning he delivered more than 20 pizzas but we do not know the exact number.

The statement doesn't mention Steve.

Trish is not mentioned either.

Vicky delivered more than 20 so she was paid a bonus.

16. B: There are 48 students where 23 are boys and 25 are girls altogether in 6F and 6S. Since there are 11 girls in 6S, there are 14 girls in 6F.

17. D: If the probability of taking out a red cube is $\frac{1}{5}$, then there must be 2 red and 8 black cubes in the bag. The total number of cubes is 10; it cannot be 5 or 15.

After 1 red is taken out, there will be 1 red and 8 black cubes left.

Hence, the probability of black in the second draw is 8 out 9.

18. A: Fare = \$2 + \$15 = \$17

Fuel cost = 15 km $\times$ 2 $\times$ \$0.40 = \$12

Profit = \$17 – \$12 = \$5

19. C: Using advertisements in social media influences what people think, which can affect the election results.

20. C: Buying a puppy does not guarantee that the owner will not change their mind about them.

21. D: Some people might get A for the report and score 100% in the test so there may be more than 3 people who got 'A' in the most recent school report.

22. D: Some people may appear as picky eaters, when in fact they have allergies that can harm them.

23. B: All three cases have a minimum of 6 letters.

• – • / – • • / – – – / • / – • • / • – •

• – • / – • • / – – / • / – • • / • – •

• – • / – • • / – – / • / – • • / • – •

24. B: The base of the solid is a small circle, while the uppermost face is a larger circle.

When viewed from the direction of the arrow, the two faces are connected in a straight line.

25. C: The total number of seats in A and D parties combined should be equal to the total number of seats in B and C parties combined.

The angle sum of C and D is larger than 90° , so the total number of seats in C and D parties combined should be greater than 30.

26. A:

50c	\$1	\$2
	1	2
2		2
	3	1
2	2	1
2	4	

27. C: cake + ice < balloons + food

cake + balloons < ice + food

The budget for cake cannot be more than the budget for food.

28. C: If there is a seventh floor, there will be a lift. Also, buildings with more than three floors will have lifts on the second floor.

29. B: A 4.3 ml to 4.5 ml = 0.2 ml increase, B 2.9 ml to 3.8 ml = 0.9 ml increase, C 3.8 ml to 4.5 ml = 0.7 ml increase, D 4.5 ml to 4.5 ml = no increase.

30. D: Lesley 10:51 scored 5 points.

George 11:39 scored 2 points.

Anyone within 90 seconds from 11:39, which is 13:09, scores 1 point. Those are Arthur, Tammy, Vicky, Sana, Carol, Adam and Pippa.

31. B: 4 day pass for 2, 3, 4 = \$11

1 day pass for 6 = \$6

7 day pass for 10, 11, 12, 14, 15, 16 = \$22

32. B: The highest combined score for 2 tests = 85 (Student B)

The second highest combined score = 77 (Student I)

If the second highest person scores 50 (the highest score possible in one test), the total will be 127.

$127 - 85 = 42$ so student B needs to get 43 to secure first place.

33. D: The light is not flashing right now because the battery could have already run out or it could start flashing soon.

If the light is on continuously, it means that the battery has not run out.

34. D: Because Rosa's dogs are mixed, it cannot be concluded that they are moderate barkers.

35. B: One of Mon or Thu is free. The total is \$7.20.

Mon	Tue	Wed	Thu	Fri
\$1.20 Free	\$2.50 + \$0.50	\$1.50	\$1.20	\$1.50

36. D: Every option is possible except if Tiffany wakes up late but is not late for work. If Tiffany wakes up late, then she will miss the bus then be late to work. Tiffany setting up an alarm does not guarantee that she will wake up on time.
37. A: Three quarters of the way between A and M is J.
J is halfway between D and P.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

38. D: An experienced acrobat does not need to sing or dance. Ross must not have filled in the form.
39. A: square = 25, circle = 37.5, triangle = 50
40. D: The age group of people purchasing games can be changing over time.

TEST 2

- C: Wearing contact lenses can cause eye infections, which can lead to blindness.
- C: We already know that the poodle was cute. If James did not choose the dog, it must be because it does not shed a lot but has low liveliness OR if the dog sheds a lot and not very lively. Telling if the poodle sheds a lot or does not shed a lot is not enough information. C shows that the poodle does not shed a lot and is not very lively.
- D: The three numbers are 29, 40, 44.
The second number is formed by $29 + (2 + 9) = 40$.
The Third number is formed by $40 + (4 + 0) = 44$.
- D: If Sarah likes drawing, she likes reading. If she likes reading, she also likes singing but does not like dancing. It is not the same if it goes the other way around.
- A: Bears hibernate (sleep for a long time) during the winter, when it is cold.
- C: $J + F > R + S$ and $J + S > R + F$ indicate that the Rose is \$29, the cheapest.
Thus, the new price can be evaluated as below.

Fragrance	Cost (\$)
Jasmine	56
Rose	29
Fruity	82 or 63
Sandalwood	63 or 82

7. B: The passage is arguing that the traditional methods of healing are more important than science. However, B demonstrates that modern science has helped people live longer, which clearly weakens the passage's argument.
8. A: Walking 6 floors down takes 120 seconds.
Going up the lift 3 floors and down 8 floors (floor 9 to floor 1) and 4 seconds from floor 1 to the ground = $16 \times 11 + 4 = 180$ seconds.
The difference is $180 - 120 = 60$ seconds.
9. A: The passage is arguing that airlines should pay a tax on aviation fuel because this will help to resolve the pollution problems caused by air travel. If airfares are increased and fewer people fly, there will be decrease in air pollution. This will directly strengthen the passage's argument.
10. D: $315 - 23 = 303$ sheep had lambs.
 256 sheep had 1 lamb each = 256
 $303 - 256 = 47$ sheep had 2 lambs
 $47 \times 2 = 94$ lambs
 $256 + 94 = 350$ lambs
11. D: C and D satisfies most of the descriptions. However, there is no mention of a separate toilet, so D is the most correct answer.
12. B: The passage is arguing that workers working in a three-shift system will lose weight because they will experience a severe decline in their appetite (feeling less hungry). Since eating less food will lead to weight loss, B strengthens the passage's argument.
13. B: If the gold is in:
Chest 1: 3 statements are true.
Chest 2: 1 of the statement is true.
Chest 3: 0 statements are true.
Chest 4: 3 statements are true.
14. A: When we read all the 3 results from the research, we can conclude that, on average, the Italian speakers are the smartest, then the Latin speakers and then the Spanish speakers. However, some students most fluent in Spanish are smarter than the students most fluent in Italian. This shows that the range (the difference between the highest and the lowest) of intelligence of Spanish speakers is the greatest.
15. D: We cannot determine where point S lies between P and R.
16. D: There are two points that must be checked:
– If the stain was removed, and if Quick-Stain was used.
– If the results are No and Yes respectively, it can debunk the claim.
17. C: If you take away X and Y from 900, you will get the number of people who were not students.

18. B: There are 19 days with one training and 9 days with two trainings. 3 days with no training. There are in total $19 + 18 = 37$ trainings.

Sun	Mon	Tue	Wed	Thu	Fri	Sat
		1 ✓✓	2	3	4	5
6	7 ✓✓	8 ✓✓	9	10	11	12
13	14 ✓✓	15 ✓✓	16	17	18	19
20	21 ✓✓	22 ✓✓	23	24	25	26
27	28 ✓✓	29 ✓✓	30	31		

19. B: 4 books from shop A = $\$16 + \$16 + \$8 + \$8 = \$48$

4 books from shop B = $\$14 \times 3 = \42

4 books from shop C = $\$72 - \$14.40 = \$57.60$

4 books from shop D = $\$15 \times 4 = \60

20. A: Adding the same amount to each side maintains the balance.
21. D: Driving slowly will allow drivers to react quickly to dangerous situations in school zones. This can apply to all the drivers. Therefore, D most strengthens the passage's argument.
22. C: $7 \text{ owners} \times 2 = 14 \text{ legs}$
 $58 - 14 = 44 \text{ legs}$
 $44 \div 4 = 11 \text{ dogs}$
 $11 - 7 = 4$
 Therefore, 4 owners have 2 dogs, and 3 owners have 1 dog.
23. A: The main idea of the passage is that there are a lot of difficulties with travelling to space. Since A is highlighting the positive side of space exploration, it most weakens the passage's argument.

24. D:

	Tennis	Origami	Castle	Swimming	Climbing
Adrian	✓				✓
Issac	✓				✓
Sarah		✓		✓	✓
Lucas		✓			
Iris	✓		✓		✓

25. C:

Mon	Tue	Wed	Thu	Fri	Sat	Sun
Rain	Dry	Rain	Dry	Rain	Dry	Rain

Saturday and Tuesday must be dry because there cannot be three rainy days in a row. This means that Wednesday has to be rainy because it is dry on Tuesday and Thursday; and Friday has to be rainy because it is dry on Thursday and Saturday.

26. D: 400 red shoes were sold.
27. C: One person could have broken multiple records, so there might be less than six record-breakers.
28. C:

Time taken to make the dish: from least to most time	Taste: from best to worst
Frankie	Gordon
Gordon or Randy	John
Gordon or Randy	Frankie
John	Randy
Teresa	Teresa

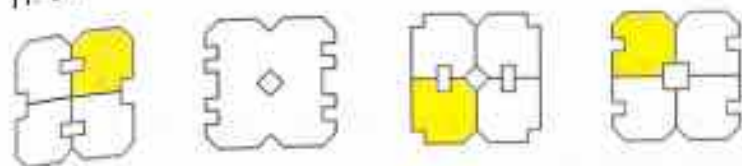
29. C: Harry can have both traits and not sprint, but he must have both traits in order to sprint. Not having one of them means he cannot sprint.
30. A: The passage's main idea is that it is better to have a shower at night because it is more hygienic. A is the only one that suggests another benefit of showering at night. Therefore, A best supports the passage's argument.
31. C: The main idea of the passage is that it is more efficient for everyone to donate organs by automatic consents, rather than volunteering. Commodifying human organs means treating human organs as products or materials, which is clearly a negative impact of a such system. Therefore, C most weakens the passage's argument.
32. D: Having the same diet will not decrease the rate of the other cancer. Both types of cancer are severe.
33. B: First two letters = $5 \times 4 = 20$ ways.
Last two digits = 6 ways.
(1, 1) (1, 2) (1, 3) (2, 1) (2, 2) (2, 3) (3, 1) (3, 2) (3, 3)
 $20 \times 6 = 120$ ways.
34. D: 
35. C: Sarah needs to use the photocopying option to copy her pamphlets. She needs to pay $2 \times 50\text{¢} \times 60 = 6000\text{¢} = \60 . After the discount, the total price is \$54.
36. D: If Penelope cannot sing, her other talents will not allow her to audition for the contest.
37. D: Clare can have both experience and money, but decide not to cook. However, she must have both traits in order to cook. Not having one trait means she cannot cook.
38. B: We are not given any information on individual games, so Billy's statement cannot be verified. He could have had less home runs than Joe in the third game and still be the overall winner of the home run race.

39. **B:** A and D are not relevant. The students who got 80% may have also gotten the last two questions correct, which in turn means there is an overlap in the numbers. Thus, there can be less than 15 people who will go through to the next stage.
40. **A:** The passage is arguing that people should use more cash than card. A, C and D provide benefits of using card. However, A is the only option that makes a counter-argument to the passage as it demonstrates the benefit of your locations being tracked. Therefore, it best weakens the passage's argument.

TEST 3

- C:** Lower levels of human activity have resulted in a rise in the animal population.
- D:** The slow train took 3 hours 30 minutes.
The express train took 1 hour 10 minutes so it departed from Central Station at 11:50 am.
- C:** From oldest to youngest:
David – flute, volleyball
Emily – drums, soccer
Abby – violin, swimming
Charlie – piano, track
Brian – trumpet, basketball
- C:** The main idea of this passage is that rain affects the operation of computers. However, if the computer stops working during all weather conditions, there are most likely other factors that contribute to the operation of computers. Thus, it best weakens the passage's argument.
- B:** The main argument of this passage is that children should aim high because it can be good motivation. However, we can logically predict that aiming high does not always guarantee achieving goals. If failing to achieve goals can result in disappointment, it definitely weakens the argument.
- B:** 300 g of flour makes 24 cupcakes.
500 g of sugar makes 40 cupcakes.
300 g of butter makes 24 cupcakes.
5 eggs makes 20 cupcakes.
With 5 eggs, 20 cupcakes can be made regardless of the amount of other ingredients.
- D:** Isabella had not joined the team before so in order to join, she needed to learn a dance choreographed by the coach in one week and audition with that dance. She must not have learned the choreography correctly.

8. D: We cannot conclude that every student plays at least one instrument because there can be students who do not play either of the instruments.
The number of players for each instrument depends on how many do not play either.
But at least $(70 + 35) - 100 = 5$ play both the piano and violin.
70 players of piano are twice the 35 players of violin.
It is possible for all 35 violin players to play the piano, so, at most, 35 students can play both piano and violin.
9. C: $A10 \times 2 + B10 \times 3 + C3 \times 5 = 65$ marks; The largest number of questions correct & fail = 23.
 $B7 \times 3 + C10 \times 5 = 71$ marks; The smallest number of questions correct & pass = 17.
10. A: $40 \times \$7 - 35 \times \$8 = 0$
11. B:



12. C: 1 to 3: 4 day pass \$70
5: 1 day pass \$35
10 to 16: 7 day pass \$140
Total \$245
13. A: Let the total number of fruits = X
 $X - 4$ = number of oranges
 $X - 4$ = number of apples
 $X - 4$ = number of bananas
 $X + X + X - 4 - 4 - 4$ = total number of fruits = X
 $X + X - 12 = 0$
 $X = 6$
There are 2 oranges, 2 apples and 2 bananas (a total of 6) in the basket. Hence, all but four are oranges, all but four are apples and all but four are bananas.
14. B: Total of Nell = 170, Cindy = 178 so Cindy has the highest total.
15. C: Christine starts dance class but may not participate in the talent show.
Christine starts dance class but may not win the trophy nor the medal.
Christine cannot win both the trophy and a medal since medal is the second prize.
16. D: The teacher might be upset with Sally for something else other than her not submitting the note.
17. D: Ratio of Adults : Children = $6 : 24 = 1 : 4$
Ratio of Boys : Girls = $16 : 8 = 2 : 1$

18. **C:** The argument emphasises that social media is linked to emotional health in teens. Less technology and more social interactions can improve mental health.
19. **D:** To gain the maximum efficiency whilst meeting all the conditions, one possible combination of tasks is for person 1 to do Tasks 4, 5, and 8 for a total of 28 minutes, person 2 to do Tasks 2, 6 and 7 for a total of 29 minutes and person 3 to do Tasks 1 and 3 for a total of 27 minutes. As person 2 takes the longest at 29 minutes, the shortest time for the tasks to collectively be completed is 29 minutes. Another combination of tasks with a completion time of 29 minutes is Tasks 2, 3, 4, Tasks 1, 5, 7 and Tasks 6, 8.
20. **B:** D F H A = 4681
 C G H B = 3782
 B E C G = 2537
 H G D A = 8741
 Only CGHB starts with an odd number.
21. **C:** If Sarah had 2 oranges, she must have 3 apples in order to have five fruits. This means that Emily has 3 oranges and 2 apples.
22. **A:** The argument is saying that valuable education is learning about life, not preparing for exams. So "Education for life is of greater value than passing exams" strengthens the argument.
23. **B:** The argument says that living in an apartment is better than living in a house. The statement against living in an apartment weakens the argument.
24. **B:**

Supplier	Total Price (\$)
<i>Koles</i>	31 500
<i>Woolly</i>	23 750
<i>Baldy</i>	27 000
<i>Lohay</i>	32 000

25. **B:** The phone receives an alert only when the car is unlocked.
26. **A:** 3 trips with 10 cars + 4 trips with 6 trucks = 30 + 24 = 54 cars and trucks.
27. **A:** Brand 3 was omitted, as it is an outlier in the data – the other four values are roughly similar in size, as demonstrated on the graph.
28. **D:** Derek has the greatest increase from Year 2 to Year 3.
29. **B:** The argument supports the vegan diet, which keeps blood sugar levels low. Other options do weaken the argument, but "Blood sugar levels that are too low can be fatal" is directly against the argument, which most weakens the argument.

30. C: If a person likes teaching, that person likes preparing and cleaning up, too. However, this does not mean that if that person likes preparing, they must like teaching. It is not the same if it goes the other way around.
If Sarah likes teaching, she likes preparing and cleaning up, but she does not like organising.
31. D: There could be other people with red hair and freckles besides Harry.
32. B: The passage supports eating chocolate in the morning. Statement B is against eating chocolate as it has a negative effect on the body. Statement A indicates that eating chocolate is harmful only if it goes over the recommended daily intakes. Therefore, Statement B most weakens the argument.
33. D: As tryouts happen every year, everyone has to pass the tryout each year. Darren is not guaranteed to be chosen for the team next year.
34. B: The tallest tree is an oak tree which is more than five metres tall, the tallest height of a pine tree.
35. D: As this is code to word, the rule has to be reversed.
J → J
N → 1 letter after → O
W → 2 letters after → Y
L → 3 letters after → O
Q → 4 letters after → S
N → 5 letters after → S
36. D: Nine missing tiles are as follows and central tile is highlighted.
Option D is the correct tile which needs to be rotated.



37. A: Feb and Aug are more than twice of most of other months.
Feb is less than half of Jan.
Aug is similar to Feb.
Oct, Nov and Dec are similar.
38. B: Only the average of BOOM is needed to compare as the average of the competitors can be determined from the information provided.
39. C: Since their total was different, there could be many combinations of scores.
40. D: One large triangle is missing.

Mastering Thinking Skills

TEST 4

1. D: Susan cannot play netball if there is swelling present.

2. D: 3-day pass for 1, 2, 3 = \$70

Either 3-day pass or 2 single day passes for 5 and 6 = \$70

7-day pass for 10, 11, 12, 14, 15, 16 = \$140

Total \$280

3. A: The fraction of beads found = $\frac{1}{4} + \frac{1}{3} + \frac{1}{5} + \frac{1}{6} = \frac{15+20+12+10}{60} = \frac{57}{60}$

The fraction that is never found = $1 - \frac{57}{60} = \frac{3}{60} = \frac{1}{20}$

$\frac{1}{20} = 5$ beads.

The total number was $20 \times 5 = 100$.

4. A: Everyone's total score can be determined without the missing information.

5. C: $3 \times \text{Gas} = \$15$

$1 \times \text{Tent} = \$15$

He cannot spend his entire budget if he buys any pan.

6. D: A person who can't dance still has a chance to be in the musical. The professional dancers do not have any influence on Terry's chance of being in the musical.

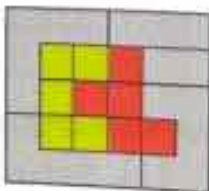
7. C: The passage is arguing that Vitamin E found in fruits and vegetables is more effective than Vitamin E in capsules. Statement C clearly shows the benefit of Vitamin E found in fruits, which most strengthens the passage's argument.

8. D: When Max is successful, he will receive \$5 from the other two players, so he will have \$30. When Alex is successful, Max needs to give him \$5. So Max will have \$25. When Kevin doesn't succeed, Kevin needs to give the other two players \$5 each. So Max will have \$30.

9. B: If someone plays tennis, it is certain they also play soccer and baseball.

10. D: Sam is arguing that carrying fire extinguishers in the car can reduce the number of car fires. However, Statement D is emphasising that carrying a fire extinguisher does not have any effect on the possibility of a car fire. Logically, carrying a fire extinguisher may reduce the damage done by a car fire but it will not stop a car fire from happening. Thus, Statement D most weakens Sam's argument.

11. D:



12. A:

	Lucy's Home	Newtown station	Midtown station	Martha's home
Arrive		10:15 am	11:35 am	11:52 am
Depart	10:05 am	10:15 am	11:40 am	

13. C: $9 \text{ cars} \times 4 \text{ wheels} = 36 \text{ wheels}$

$$60 - 36 = 24 \text{ wheels}$$

$$24 \div 2 = 12 \text{ bicycles}$$

$$12 - 9 = 3$$

Therefore, 3 cars were carrying 2 bicycles, and 6 cars were carrying 1 bicycle.

14. D: Harry was the first to finish.

Fastest to slowest	Number of hurdles knocked over
Harry	3
Nathan	2
Gina/Kenny	Gina: 1 or 0
Gina/Kenny	Kenny: unknown
Aiden	0

15. A: There are two points that must be checked: if the student is late to the class and was full at recess. If the student was not full at recess or was not late, then (b) and (d) are not needed.

16. D: Cost of rides individually:

$$\$15 + \$20 + \$10 + \$15 + \$10 + \$20 + \$15 + \$15 + \$10 + \$10 = \$140$$

$$\$140 - \$120 = \$20$$

17. C: Fruit pies must be filled with various types of fruits to be considered for the prize.

18. B: The complete diagrams for A, C, and D are shown below respectively.



19. C: The biggest sector of the pie chart is more than double the size of the second biggest and the second biggest is double the size of the smallest. The data that fulfils these specifications are 108, 43 and 22. This means the number for Mac Port was left out.

20. C: According to

$$S + C < B + D \text{ and}$$

$$S + D < B + C, \text{ the biscuits are the most expensive } (\$32).$$

21. C: The main idea of the passage is that laughter promotes healthy emotional changes in the body. Statement C is the only option that highlights the emotional benefit of laughter, which clearly strengthens the passage's argument.

22. C: It is certain that Tuesday had the most ice cream sales and Thursday had the second most. We also know that Monday had more ice cream sales than Friday. This means that ice creams sales on Friday cannot be the third highest.
23. B: In order for Holly to be certain she would be moving onto the next stage, she needs to have scored at least 20 on Test 2 (the pass mark), and 14 on Test 1 (one less than the pass mark). Any score lower than 34 would not be a guarantee to proceed.
24. A: The average of student 4 is 54. Student 3 has the highest average and will get the prize.
25. B: $3h + 3 = 1h$ of songs to be played.
In order to find the minimum number of different songs, we have to assume that every song lasts 4 minutes.
 $60 \text{ min} \div 4 \text{ min} = 15 \text{ songs}$.

26. C:

	Liam	Mary	Ben
Start	\$30	\$30	\$30
Liam rolls a 1	\$28	\$31	\$31
Mary rolls a 3	\$31	\$25	\$34
Ben rolls a 4	\$27	\$21	\$42

27. D: Everyone who is good at running happens to be able to play tennis, but not necessarily the other way around.
28. C: If Chest 1 is true, then Chest 4 is also true.
If Chest 2 is true, then there is a chance that Chest 4 is true.
If Chest 3 is true, then all other statements become false.
If Chest 4 is true, then Chest 1 is also true.

29. B:

Tilly	Charlie	Sylvia	Nick	Peter
Even	Even	Odd	Odd	Odd
4/6/8	4/6/8	7	3	9

30. B: Shop A: $(32 + 16) \times 3 = 48 \times 3 = \144
Shop B: $33 \times 4 = \$132$
Shop C: $(30 \times 6) \times 0.85 = \153
Shop D: $24 \times 6 = \$144$

31. C: $7 \text{ cups} = 6 + 1$

$$42 \text{ cups} = 6 \times 7$$

$$55 \text{ cups} = 6 \times 9 + 1$$

$$(6), (6 \times 7), (6 \times 9) = 17 \times 6 \text{ at a time}$$

$$(1), (1) = 2 \times 1 \text{ at a time}$$

32. D.

	Dennis	Bob	Coleman	Ashley	Emma
Hobby	Taking pictures	Swimming	Reading	Watching movies	Computer games
Pet	Snake	Spider	Goldfish	Dog	Cat

33. C: There is no rule stopping one student from becoming student of the month more than once.

34. B: Changeable letters = S, o, o, s
 $2 \times 2 \times 2 \times 2 = 16$

35. C: Possibility 1: Sam, Alice, Paul, Dara
 Possibility 2: Dara, Paul, Alice, Sam

36. B: The main idea of the passage is that social media users can be misinformed because there is no way of determining the accuracy of the information. If there is no program to analyse content, there is no way of determining the accuracy of information posted on Twitter, which is an example of social media. Therefore, Statement B most strengthens the passage's argument.

37. A: The possible number of pancakes that a person can make with each ingredient:

	Adam	Beckie	Cynthia	Douglas
Flour	21	18	18	21
Eggs	24	18	24	18
Milk	20	16	24	22

The lowest number of pancakes from each column indicates the number of pancakes that a person can make. Adam can make the most pancakes.

38. C: Catch a ferry at Chelsea Harbour 07:55 Westbound. Get off at Putney at 08:15. Take the ferry at Putney Eastbound at 08:40 and get off at Blackfriars at 09:35. Take a ferry from Blackfriars at 09:50 Westbound and get off at Chelsea Harbour at 10:25.

39. D: Top view of shape D.

40. C: The main idea of this passage is that the bicycle is a better alternative to the electric car because it is more cost efficient. Statement C is the only option that highlights the disadvantage of using the bicycles, thus it most weakens the passage's argument.